Benchmarking LLM Integration Strategies for MCDA-Assisted Household Energy Decision Support

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Abstract

The residential sector accounts for 18% of global CO_2 emissions [1], yet most people rely on biased decision making rather than tailored choices. Multi-Attribute Value Theory (MAVT) offers a rigorous method of Multi-Criteria Decision Analysis (MCDA) to reduce emissions, but building MCDAs daily requires expertise that limits household adoption. This study evaluated whether LLMs can bridge that gap by comparing three architectures: Direct LLM Scoring ($\mathcal{A}_D$), Example-Guided LLM Scoring ($\mathcal{A}_E$), and LLM-Parameterized Reference Scoring ($\mathcal{A}_H$), which extracts engineering parameters for a deterministic physics calculator. Ground truth calculators were built to benchmark MAVT scores across HVAC, appliance, and shower decisions. $\mathcal{A}_H$ achieved a Kendall's tau of 0.902, MAE of 0.051, and 91.7% Top-1 accuracy across 195 scenarios. $\mathcal{A}_E$ scored 0.339, 0.138, and 51.7%; $\mathcal{A}_D$ scored 0.095, 0.219, and 35.4%.

Keywords: Large language models, Multi-criteria decision analysis, Multi-Attribute Value Theory, Household energy, Decision support systems, Retrieval-Augmented Generation, Sustainability

1. Introduction

The average U.S. household emits roughly 8,744 pounds of CO_2e and consumes about 10,791 kWh of electricity each year [2, 3]. Globally, residential energy consumption has grown at roughly 1% per year since 2000, driven by increasing household sizes and comfort demands in emerging economies [1]. However, there is more than enough evidence that this could be much further optimized to reduce cost to families and environmental impact [4, 5].

Tools that combine optimization, behavioral insight, and decision support can close part of that gap, but only if they are usable without specialized training. Certain frameworks, such as Multi-Criteria Decision Analysis (MCDA), can be used to choose an optimal balance of factors that matter to users [6]. Unfortunately, these are extremely taxing to create oneself accurately even as an expert, and they require extensive time which makes using them by themselves impractical for household decisions.

However, it has been proposed in the past that LLMs could be used to automate this process, thus making it practical for household decisions. Without knowing how accurate these LLMs are as experts, it is unclear whether this would be a beneficial or detrimental idea.

Modern LLMs carry general knowledge about the energy use and carbon intensity of common products and behaviours, and many systems can retrieve additional information at inference time. They have not, however, been tested extensively for their abilities as MCDA designers. Additionally, the amount of assistance needed to improve their designer ability to a reasonably accurate level has not been tested either. This study addresses that gap by benchmarking the three architectures across three household energy decision types, holding the model and the MAVT weighting constant for each architecture.

We aimed to answer the following question: Which LLM integration architecture most accurately acts as a replacement for an MCDA expert for rapid decision making? Then, How consistently does the architecture ranking hold across models of varying capability?

2. Literature Review

2.1. *Household Energy Decision-Making and Adoption Barriers*

When transportation, food, and goods are included, the per-capita footprints rise to about 17.9 tons of CO_2e and 81,767 kWh of total annual energy [7, 8]. The U.S. Department of Energy estimates that households can save \$200–\$400 annually by adopting ENERGY STAR practices [5, 9]. Globally, residential buildings account for 18% of CO_2 emissions [1]; in the United States, the residential sector accounts for roughly 20% of energy-related CO_2 emissions [10]. Dietz et al. (2009) estimated that household behavioral changes could reduce U.S. emissions by 123 MtC (approximately 451 $MtCO_2$e) annually [4]. HVAC systems alone account for roughly a third of global residential energy consumption [1]. Figure 1 tracks Pennsylvania residential emissions from 2005 to 2022.

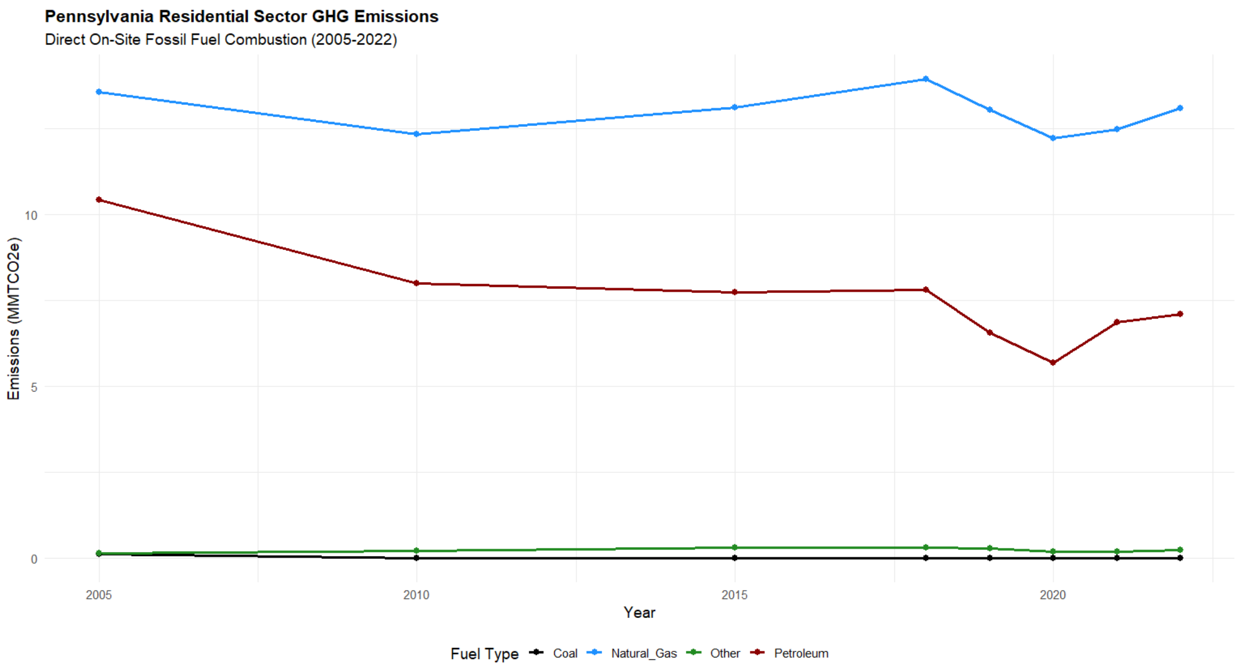

Figure 1: Residential emissions, 2005–2022 (EIA, 2022) [10, 11].

Attari et al. (2010) found that consumers undervalue energy use and savings by a factor of 2.8 on average, systematically underestimating high-energy activities while overestimating low-energy ones [12]. Thus, often the "usual setting" for most activities is based off of incorrect assumptions, whether this is a temperature setpoint, an appliance run time, or shower time.

Multi-criteria decision analysis (MCDA) can optimize complex decisions with multiple interacting variables such as household energy choices [6]. However, constructing the decision matrix is time-consuming even for experts [13, 6]. Huang and Burgherr (2024) built a streamlined MCDA calculator that compresses several accepted methods into a single workflow [14], yet the user still needs enough familiarity with MCDA construction to frame the alternatives, criteria, and weights, which keeps the tool out of routine household use. Constructing an MCDA framework for a specific decision requires methodological knowledge—choice of weighting scheme, value function form, normalization, rubric design, and domain familiarity—all of which impose barriers that limit household adoption. While the average U.S. household consumes 10,791 kWh annually, this consumption is not evenly distributed. Space heating accounts for approximately 43% of residential energy use, followed by appliances (13%), water heating (12%), and air conditioning (8%) [10]. This distribution highlights key intervention points for emissions reduction. National reform of certain key household practices could reduce national emissions by 123 million metric tonnes of carbon (MtC) per year. As of 2005, residential emissions accounted for 38% of overall U.S. CO_2 emissions, totaling 626

MtC. Additionally, these changes would result in little to no reduction in satisfaction, a necessary factor in widespread implementation [4].

Figure 2 reproduces the Dietz et al. display of behavioral plasticity, net carbon effect, and the Reasonably Achievable Emission Reduction (RAER) per action.

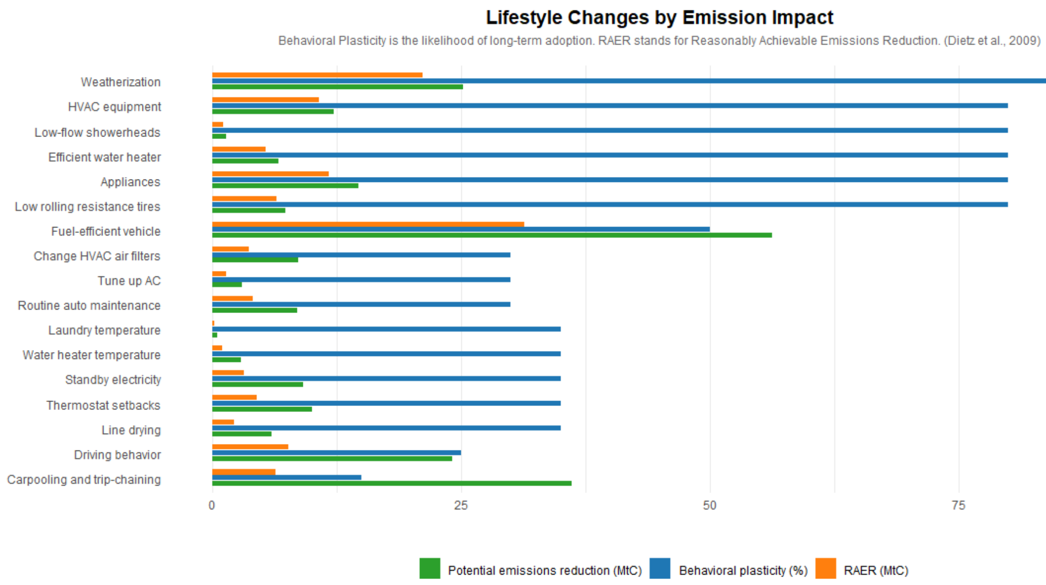

Figure 2: Household behavior changes and their emissions impact (after Dietz et al., 2009 [4]).

Dietz et al. [4] also demonstrated that household-level behavioral changes, if widely adopted, could cut U.S. emissions by 123 MtC annually. However, concerns for behavioral plasticity—how likely people are to implement each action based on financial and social incentives—remain an important factor [4]. Actions such as installing programmable thermostats demonstrate high emissions reductions and adoption likelihood, whereas vehicle upgrades show reductions but low likelihood. More often than not, individuals pointed to curtailment rather than implementing new products for energy efficiency in their households [12]. Thus, targeting simple, heuristic actions is likely to be more effective at this scale than recommending car or HVAC replacements, even if those may be more impactful.

Another problem for MCDA-style tools at the household level is the workflow tax they impose on users. Even when the benefits are clear, switching from a routine choice to a tool adds time and cognitive cost, making it less likely that something like that could persist long term. [15, 16, 17]. Most household decisions are made by repeating a relatively random choice, rather that actually analyzing the environment before beginning.

Heuristics are useful in everyday life, but under complex or high-stakes situations, they can lead to errors; multiple studies have found that under time pressure, people default to ”satisficing” (choosing the first option that meets minimum criteria) or using dominantpast behaviors, even if they’re not ideal [18]. Household energy decisions typically occur like this: short consideration windows, competing demands, and incomplete information. Tools that require sustained attention or structured inputs—even if it was completely accurate, such as a calculator built for the situation—are unlikely to influence behavior under those conditions.

2.2. LLMs as Numeric Reasoners and Scorers

Cobbe et al. (2021) introduced the GSM8K benchmark of grade-school arithmetic word problems and found that even large Transformers fall well short of human accuracy once a problem needs more than a few chained steps, with errors compounding rather than cancelling [19]. Imani et al. (2023) approach the same problem from the fix side, building MathPrompter on the observation that LLMs "often provide incorrect answers" on arithmetic benchmarks unless cross-checked against multiple generated solution paths [20]. As such, expecting LLMs to solely be responsible for and accurately understand how competing (and often similar) alternatives can have one clear winner alone is therefore unreliable without external grounding.

The original uses of RAG is injecting external facts into an LLM's context to reduce hallucination in open-domain QA [21]. An MCDA scoring task is closer to Case-Based Reasoning, which treats problem-solving as retrieving similar prior cases and reusing their recorded solutions [22]. Wiratunga et al. (2024) build CBR-RAG, where retrieved legal cases act as structured exemplars that shape an LLM's reasoning rather than supply facts [23]; Minor and Kaucher (2024) get a similar result, finding that retrieved example explanations used as in-context demonstrations improve LLM-generated outputs over zero-shot prompting [24]. Thus, providing pre-scored expert scenarios, which essentially can provide more evidence and nudge the LLM in the correct dimension, could reasonably increase accuracy. Additionally, creating databases with graded expert examples is not an infeasible task for large scale use; the US Energy Information Administration's Residential Energy Consumption Survey, for example, has produced nationally representative household energy profiles since 2005 through standardized survey instruments, and comparable instruments exist in other nations [1].

However, both of these implementations have their fundamental flaws; Wang et al. (2024) show that LLMs exhibit positional bias when scoring, ranking the first response higher simply because of its placement [25]. Their conclusion that LLMs are "not fair evaluators" in default zero-shot configurations [25] motivates our study rather than contradicts it. Firstly, the models used in the tests were relatively weak compared to newer models in reasoning. Thus, newer testing on more powerful models can reveal new results.

Li et al. (2026) document systematic scoring-bias distortions in LLM-as-a-judge pipelines that persist even in advanced models [26], and Zhu et al. (2026) trace most of the cross-run score variance on benchmarks like MT-Bench to judge identity rather than response quality [27]. Johnson and Zhang (2024) find GPT-4o essay scores clustering below human ratings with less variance—a central-tendency bias where the model avoids extreme scores [28]—and Poličar et al. (2025) show that different LLM graders carry distinct, model-specific leniency profiles relative to human TAs [29]. Based on this prior work, we would predict that $\mathcal{A}_{\mathrm{D}}$ exhibits a clean failure mode: the four criterion scores for all three alternatives cluster into a narrow middle band, missing the sharp, non-linear penalties in the ground-truth value functions—the shower calculator's tank-capacity cutoff, the budget penalty's exponential tier—and dragging down Kendall's tau even when scores are directionally reasonable.

An alternative to having the LLM score directly is to have it estimate the missing engineering parameters and let a deterministic calculator handle the scoring. This LLM-parameterized approach preserves the transparency of equation-based evaluation while leveraging the LLM's pattern-matching ability to infer quantitative values from natural-language context. The tradeoff is narrower applicability: a calculator must be built for each decision domain, which is not always practical for quick-turnaround or one-off evaluations. The study tests this tradeoff directly through $\mathcal{A}_{\mathrm{H}}$, which pairs single-call LLM parameter extraction with the three ground-truth calculators.

Despite extensive work on LLM numeracy, retrieval-augmented generation, and LLM-as-judge scoring, no prior study has rigorously compared these integration strategies against a deterministic, physics-based ground truth for household energy decisions. The present study fills this gap by benchmarking three LLM-MCDA architectures—$\mathcal{A}_{\mathrm{D}}$, $\mathcal{A}_{\mathrm{E}}$, and $\mathcal{A}_{\mathrm{H}}$—across 195 test scenarios spanning HVAC, appliance, and shower decisions, using identical MAVT weighting and model access so that the integration strategy is the sole variable under test.

3. Methodology

Table A.15 in Appendix A lists the notation used throughout this paper.

3.1. Decision Types and Scenario Corpus

Three decision types were selected for their high behavioral plasticity and RAER: HVAC (thermostat setpoints), Appliance (time-of-use scheduling), and Shower (duration). HVAC was prioritized because space heating and cooling account for approximately a third of global residential energy consumption and represent the fastest-growing end-use as household incomes rise [1].

The scenario corpus has two pools. The test set contains 195 scenarios (70 HVAC, 65 Appliance, and 60 Shower); each scenario presents three alternatives scored independently by each architecture and by the ground-truth calculator.

The *RAG corpus* (Appendix E) contains an additional 90 scenarios (35 HVAC, 35 Appliance, and 20 Shower) that populate the retrieval index used by the $\mathcal{A}_E$ architecture; RAG scenarios are never used for evaluation [21].

The split provides a 2.17:1 test-to-RAG ratio sufficient for retrieval coverage; HVAC has more scenarios than the others (70 test + 35 RAG) due to its higher parameter dimensionality (R-value, SEER, age, occupancy, insulation tier). The same applies to Appliance decisions.

The scenario counts were chosen to ensure full coverage of the parameter space rather than to satisfy a statistical power calculation. Each decision type's scenario set was generated by crossing the categorical parameters (insulation tier, housing type, appliance type, flow-rate band) with representative numeric draws (outdoor temperature, square footage, household size, utility budget) and then auditing the resulting matrix for coverage gaps—combinations of categories that appeared in fewer than two scenarios were filled with additional draws until no parameter combination was underrepresented. This combinatorial audit was performed using a consolidated workbook that tracked parameter-frequency histograms; scenarios were redistributed from overrepresented cells to underrepresented ones using an automated rebalancing script until each categorical cross-cell contained at least one scenario. The 90 RAG scenarios were drawn from the same parameter distribution but kept fully disjoint from the test set, ensuring no test scenario had an identical parameter signature in the retrieval index.

3.2. End-to-End Workflow

Figure 3 shows the full decision-support workflow common to all three architectures. The three architectures differ only in how the scoring step is performed.

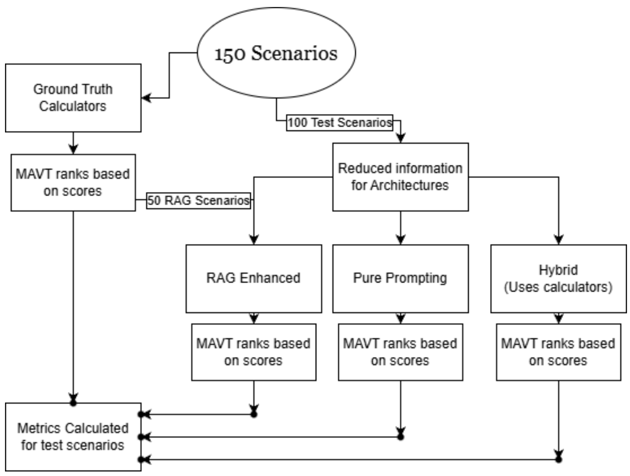

Figure 3: Decision-support workflow: scenario input → architecture-specific scoring or extraction → MAVT aggregation → ranked output.

Results are reported as five-run averages: each architecture–model combination is run independently five times with different random seeds, and metrics are computed per run then averaged.

The tables below detail the parameters given to architectures vs. calculators per decision type.

The calculator receives specific engineering values that would require professional measurement for households, and thus are unreasonable to be requested of homeowners. The architectures receive generalized version (e.g. insulation, which is a general estimate of poor/average/good rather than a specific R-value) or do not receive counterparts to some parameters at all (e.g. SEER and R Values). These parameters are chosen as they are things a household member would actually know [30].

We first built the scenarios with all parameters completely specified. This was the information that the ground truth calculators received. Then, we identified acceptable ranges for each from published standards and mapped each scenario to a representative value within that range.

Each insulation tier spans a range of R-values drawn from standard U.S. residential wall assemblies [31, 32]: Poor scenarios use R-8 to R-12 (minimal 2×4 cavity fill, below current code), Medium scenarios use R-13 to R-18 (standard 2×4 batt through 2×6 mixed), and Good scenarios use R-19 to R-24 (2×6 framing and above). The tier boundaries are chosen so that adjacent tiers differ by 15–20% in wall R-value, matching the minimum difference a homeowner

Table 1: HVAC parameters: Architectures vs. calculator

Parameter	LLMs	Calc.
Location	✓	—
Outdoor Temp	✓	✓
Home Size (sq ft)	✓	✓
Insulation (Good/Average/Poor)	✓	—
R-Value	—	✓
Household Size	✓	✓
Housing Type	✓	✓
House Age[a]	✓	✓
Utility Budget	✓	✓
SEER Rating	—	✓
Occupancy Context[b]	✓	✓
Indoor Temp (from alternative)	—	✓

[a]LLMs receive a building-age range label (e.g., "20+ Years"); the calculator uses HVAC age, which is around or less than building age, and an exact number [b] Question context provides context.

could detect by inspecting cavity depth. SEER ratings in the test corpus span 8–22, covering pre-1992 units (SEER 8–10), standard-efficiency replacements (SEER 13–14), and high-efficiency models (SEER 16–22), consistent with the ENERGY STAR Most Efficient threshold [33].

Table 2: Appliance parameters: Architectures vs. calculator.

Parameter	LLMs	Calc.
Appliance Type	(text)	✓
Baseline / Run Time	(text)	✓
Appliance Age Range	✓	—
kWh per Cycle	—	✓
TOU Period / Utility Rate	—	✓
Monthly Cycles Assumption	—	✓
Household Size	✓	✓
Housing Type	✓	✓
Location	✓	✓
Utility Budget	✓	✓

Appliance age bands (1–3, 4–6, 7–9, 10–12, 13–17, 18–22 years) map to measured kWh/cycle ranges drawn from the ENERGY STAR certified-products database, so that a "7–9 year" dishwasher maps to the same energy draw whether the true age is 7 or 9. Housing type (apartment, condo, townhouse, rowhouse, twin, single-family) determines the envelope multiplier in the HVAC load equation and the noise propagation factor in the appliance comfort model; these mappings are detailed in Section 3.6.

For the Flow Rate parameter, the scenario's numeric GPM is bucketed into three qualitative labels: `low_flow` ($\leq$2.0 GPM, including EPA WaterSense certified 1.5 GPM showerheads [34]), `standard` (2.5–3.0 GPM, at or near the Energy Policy Act of 1992 federal maximum of 2.5 GPM [34]), and `high_flow` (>3.0 GPM). The label is shown to the LLM; the calculator always receives the exact numeric GPM from the scenario.

The two sets overlap on environmental conditions so that both pipelines are reasoning about the same physical scenario. The LLM never sees the engineering parameters directly. For the $\mathcal{A}_{\mathrm{H}}$ architecture, which needs to use those parameters within its calculations, it extrapolates estimates of them from the question text and other parameters before the calculator is invoked.

3.3. MAVT Framework & Weighting Design

The Multi-Attribute Value Theorem (MAVT) allows alternatives to be compared on different criteria based on the range of values that occurs for each criterion. In our study, we chose the following weights: Environmental Impact (35%), Energy Cost (30%), Comfort (20%), and Practicality (15%), as summarized in Table 4.

Table 3: Shower parameters: Architectures vs. calculator.

Parameter	**LLMs**	**Calc.**
Location	✓	✓
Outdoor Temp	✓	✓
Inlet Water Temp	—	✓
Heater Setpoint	—	✓
Flow Rate (low_flow/standard/high_flow)	✓	—
GPM	—	✓
Duration (min)	(alt.)	✓
Household Size	✓	✓
Tank Size	—	✓
Utility Budget	✓	✓
Housing Type	✓	✓

Criterion	Weight (%)	Value Function	α	Direction
Environmental Impact	35	Linear (Eq. 2)	—	Lower raw = better
Energy Cost	30	Linear (Eq. 2)	—	Lower raw = better
Comfort	20	Logarithmic (Eq. 3)	1.5	Higher raw = better
Practicality	15	Logarithmic (Eq. 3)	1.2	Higher raw = better

Table 4: MAVT criterion specification. Weights sum to 100% and are held constant across all architectures and the ground truth calculator.

The weights are held constant across all decision types, all architectures, and the ground-truth calculator. We did this so that any difference in score is because of the inputs and value function, not because of difference in the way each criterion is perceived for the three decision types. Normally, MAVT additive form (Eq. 1) requires preferential independence among criteria [35, 36]. This means that criteria should be independent, meaning that the value (good or bad) of one criterion should have no effect on the value of another criterion.

$$s_j = \sum_{i=1}^{4} w_i \cdot v_i(x_{ij}) \tag{1}$$

In the case of our criteria, the energy cost and the environmental impact criterion are not completely independent, as they both stem from the kWh usage of an activity; The Spearman's rho between energy cost and environmental impact is 0.92. However, this is a feature of the dataset and of household emissions, and it is actually possible for the two criterion to diverge. Since environmental impact comes from actual CO_2, which varies based on the type of power plant generating the power, grid mix composition, renewable penetration, and regional interconnection flows [37], and energy cost comes from the time of day, gas peakers, demand charges, transmission congestion pricing, and utility revenue requirements [38]. While Pennsylvania does not usually have such divergence, there are other areas that do, such as Texas (ERCOT), where high daytime wind generation can drive electricity prices near zero while gas-fired backup units sustain elevated marginal emissions during evening demand peaks [39, 40].

The Environmental weight of 0.35 reflects two convergent considerations. Residential emissions account for a sizable share of national CO_2 emissions [4], so a household decision-support tool that does not give priority to environmental impact fails in its objective. The Value–belief–norm theory, which says that environmental orientation is the most consistent normative driver across pro-environmental household behaviours, also applies heavily due to the nature of the people using the tool. [41, 42, 43, 44, 45, 46, 47, 48]. Finally, entropy-based normalization of the scenario distributions assigns higher information weight to the environmental criterion, independently supporting an above-average allocation [49] (see Appendix B for the full computation).

For the average person, cost is the strongest predictor of consumer adoption. This is clear for one central reason: cost savings as shown on energy labels for consumer products were even more effective than actual energy use and carbon emissions.[50]. Weighting Energy Cost at 0.30 keeps it slightly below environmental impact while keeping its role as the criterion households actually act on. For HVAC decisions, thermostat setpoints directly determine electricity draw at the billing rate; households shown real-time cost feedback respond by reducing the difference from outside temperature whenever they can [51]. For appliance decisions, energy cost depends on the time-of-use rate at which the appliance runs; a dishwasher shifted from peak to off-peak can cut its per-cycle cost by 30–50%, making cost the criterion most responsive to scheduling changes. For shower decisions, water heating accounts for a significant share of residential energy costs and the per-shower increment is rarely visible to users, amplifying the underestimation bias found by Attari et al. and making cost-feedback interventions similarly effective. [52, 53].

Comfort is weighted at 0.20. Thermal comfort is the second driver of HVAC behavior even sometimes when it conflicts with energy savings [54, 55, 56]. The set points used for HVAC comfort were based mainly off of ASHRAE 55 [57, 58]. For appliance decisions, comfort is primarily a function of run-time delay: demand-response research finds that household willingness to accept a shift decays sharply beyond two to four hours, with dishwashers tolerated longest and dryers shortest [59, 60], and the penalty scales further with household size as dishes and laundry accumulate faster. For shower decisions, comfort peaks around the REU2016 average of 7.8 minutes [52, 53] and depends additionally on temperature adequacy, with heater setpoints below CDC Legionella thresholds carrying a comfort penalty [61].

Practicality is weighted at 0.15. In our context, practicality means the likelihood that an activity will be continued long term. While this is partially linked to the other criterion, it also requires that the tool adds minimum friction relative to the existing routine. Otherwise, adoption drops sharply, and heuristic decision-making remain the default for everyday energy choices [15, 16, 62, 63]. Penalizing alternatives that violate budget, system, or behavioral limits keeps the tool's recommendations in check to make sure that households would actually adopt it long term. The lower weight (relative to comfort) reflects that practicality is a constraint on what is reasonable rather than a primary preference dimension. Where comfort captures what a household would prefer given no constraints, practicality captures what is feasible and sustainable to adopt. A 15-minute shower may score near-optimally on comfort, but if four occupants share a 40-gallon tank, available hot water runs out before the third person showers—making that alternative infeasible regardless of comfort preference.

By requiring a score or parameter estimate for each of the four criteria, the structured framework prevents the LLM from arbitrarily selecting an alternative based on a single dominant attribute, a known failure mode of unaided LLM judgment [25].

$$v_i(x) = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{2}$$

$$v_i(x) = \frac{\ln\left(1 + \alpha \cdot \frac{x - x_{\min}}{x_{\max} - x_{\min}}\right)}{\ln(1+\alpha)} \tag{3}$$

3.4. Sensitivity and other analysis

A sensitivity analysis was conducted to test the stability of the results. Fluctuations of ± 0.05 were applied to the weight vector by perturbing each of the four criterion weights individually and redistributing the difference equally across the remaining three, plus one equal-weight scenario ($w_j = 0.25$ for all j).

Table 5: Reference ranges (5th–95th percentiles). Shower Environmental Impact is reported in gallons of water; HVAC and Appliance in lbs CO_2.

Criterion	HVAC	Appliance	Shower
Energy Cost ($)	0.38–3.29	0.025–0.71	0.14–1.14
Environmental Impact	1.96–18.04	0.288–3.643	6.0–45.0
Comfort	0.0–1.0	0.0–1.0	0.0–1.0
Practicality	0.05–1.0	0.05–1.0	0.05–1.0

3.5. Reference Ranges

The reference ranges in Table 5 are computed as the 5th–95th percentiles of the ground-truth quantity distributions over the scenario corpus (cost and emissions for HVAC and Appliance; cost and water volume for Shower), rather than extreme theoretical limits, so that normalization preserves meaningful spread across typical residential alternatives.

However, since these values were calculated from actual setpoints from Pennsylvania, they do reflect real world usage. Using dataset percentiles rather than the full theoretical physics envelope is a deliberate modeling choice [49] at the cost of truncating scores for extreme outlier scenarios. The endpoints below are the percentile values to two decimals.

For HVAC, the cost percentiles are computed over the active-conditioning alternatives only: a setpoint at or near the outdoor temperature (and the bare "Off" option) produces zero decision-period load and collapses to \$0, which would make the lower bound degenerate. Excluding those zero-load points, the realized 8-hour cost distribution at Pennsylvania's residential rate (\$0.19/kWh) [64] spans \$0.38–\$3.29; an Off or zero-load alternative still receives the top of the cost score via the decreasing-value normalization. These endpoints remain consistent with residential HVAC efficiency studies (efficient: [65]; degraded: [66]). The HVAC environmental bounds (1.96–18.04 lbs CO_2) are obtained by mapping the same cost-derived kWh envelope through PJM marginal emissions factors (0.976 off-peak, 1.041 peak lbs CO_2/kWh) [37].

For appliances, the realized per-cycle cost distribution (kWh/cycle multiplied by the resolved time-of-use rate across the six PA utilities) spans \$0.025–\$0.71, and the realized per-cycle emissions distribution (kWh/cycle through the PJM marginal factor at run time) spans 0.288–3.643 lbs CO_2. The underlying per-cycle energy endpoints remain consistent with the ENERGY STAR certified-products distribution (efficient HE washers near 0.25 kWh) and include the older non-certified electric resistance dryers [67, 68, 69, 70, 71, 72].

For showers, the realized per-shower cost distribution spans \$0.14–\$1.14. Shower energy follows the mixing-fraction physics targeting a fixed 105 °F delivery temperature, so it depends only on the rise from the mains inlet (seasonal inlet temperatures from NREL hot-water modeling [73]) to that target and is independent of heater setpoint; cost is this energy priced at the PA residential rate. Shower environmental impact is defined as water volume (gallons); the realized distribution of flow rate multiplied by duration spans 6.0–45.0 gallons at the 5th–95th percentile, with flow-rate and duration benchmarks from EPA WaterSense Appendix B [34], REU2016 [52], and The Harris Poll [53]. This was done as opposed to using simple carbon dioxide because CO_2-equivalent estimates would add a second layer of model-dependent assumptions—water-heater efficiency, local grid mix for heating energy—without improving the ranking, since the mixing-fraction physics already captures the relevant cross-alternative variation.

3.6. *Ground Truth Calculators*

Each ground-truth calculator consumes a scenario with three alternatives and returns four scores per alternative—Energy Cost (\$), Environmental Impact, Comfort, and Practicality (the latter three on 0–1)—plus the raw physical quantities for environmental impact and energy cost before value-function transformation. Energy cost is always in dollars per 8-hour decision period; Environmental Impact is in lbs CO_2 for HVAC and Appliance decisions and in gallons of water for Shower decisions.

For HVAC and Appliance decisions, environmental impact uses PJM marginal emissions factors:

$$\mathrm{CO_2} = E_{\mathrm{kWh}} \cdot \varepsilon(t), \qquad \varepsilon(t) = \begin{cases} 1.041 \text{ lbs/kWh} & \text{peak (7am–11pm)} \\ 0.976 \text{ lbs/kWh} & \text{off-peak} \end{cases} \tag{4}$$

where E_{kWh} is the decision-period energy use and $\varepsilon(t)$ is the PJM marginal emissions factor at run time t [37]. Marginal rather than average factors are used because shifting a residential load displaces or adds the generator at the margin. All three calculators apply an identical four-tier budget penalty $P(u)$ multiplicatively to the energy-cost value-function output, where $u = C_{\mathrm{monthly}}/B$ is monthly cost utilization relative to the household's stated budget B:

$$P(u) = \begin{cases} 1.0 & u < 0.80 \\ 1 - 2.5\,(u - 0.80) & 0.80 \le u < 1.0 \\ 0.5\,e^{-3(u-1.0)} & 1.0 \le u < 1.5 \\ 0 & u \ge 1.5 \end{cases} \tag{5}$$

where $P(u)$ is a unitless multiplicative penalty on the (post value-function) energy-cost score, u is monthly budget utilization, C_{monthly} is estimated monthly energy/water-related cost for the decision type, B is the household-stated monthly budget for utilities, and e is Euler's number. The no-penalty zone ($u < 0.80$) reflects mental budget safety margins [74]; the linear decline approaching the limit follows Heath & Soll's budget self-control model [75]; exponential loss aversion for overruns follows Prelec & Loewenstein [76] and Heutel [77]; and elimination beyond 150% reflects financial infeasibility [78]. Applying the penalty multiplicatively ensures extremely cheap or expensive alternatives cannot completely sway ranking on energy cost alone.

3.6.1. HVAC Calculator

Thermal load decomposes into four ASHRAE-style components. For cooling:

$$Q_{\text{cool}} = \underbrace{\frac{m \cdot A}{R}\,\Delta T}_{\text{conductive}} + \underbrace{400n + A + 800}_{\text{internal}} + \underbrace{0.15\,A \times 20}_{\text{solar}} + \underbrace{1.08\,\frac{A\,h_c\,\alpha}{60}\,\Delta T}_{\text{infiltration}} \tag{6}$$

where A is floor area (sq ft); m is a housing-type envelope multiplier (1.7 single-family, 1.6 twin/semi-detached, 1.5 townhouse/rowhouse, 1.2 apartment/condo) [79]; R is the wall insulation R-value; $\Delta T = T_{\text{out}} - T_{\text{in}}$; n is household size; $h_c = 8$ ft and $\alpha = 0.35$ ACH for ceiling height and infiltration rate [79]. The 800 BTU/hr baseline captures always-on standby loads per ACCA Manual J [79]. The heating load uses the same expression with $\Delta T = T_{\text{in}} - T_{\text{out}}$, no solar term, and internal gains subtracted rather than added.

Effective EER is estimated from the unit's rated SEER via the AHRI 210/240 quadratic [80]:

$$\text{EER} = -0.02\,\text{SEER}^2 + 1.12\,\text{SEER} \tag{7}$$

where SEER is the unit's rated seasonal energy efficiency ratio and EER is the resulting effective energy efficiency ratio used for the decision period. Age- and maintenance-driven efficiency degradation is deliberately *not* applied to the energy path; it is instead modeled as a reliability factor in the practicality score (Eq. 12), so the energy score reflects the setpoint choice rather than the system's condition. Additionally, the effect on the system in terms of energy usage will almost always be negligible for the period of consumption, thus not impacting energy use. Energy consumption over the 8-hour decision window is then:

$$E_{\text{kWh}} = \frac{Q_{\text{load}}}{\text{EER} \times 1000} \cdot 8 \cdot m_{\text{occ}} \tag{8}$$

where Q_{load} is the thermal load (BTU/hr) for the decision context and m_{occ} scales for occupancy context: 1.0 for fully occupied, 0.75 for overnight, and $1 - 0.5(h_{\text{away}}/24)$ for daytime unoccupied periods [81], where h_{away} is hours-away per 24-hour day.

Because HVAC alternatives carry no explicit run-time, the marginal emissions factor in Eq. 4 is resolved from the occupancy context by applying each pattern onto the PJM peak (7am–11pm, 16 h) and off-peak (8 h) windows, with away-hours assumed to fill the daytime peak window first:

$$\varepsilon_{\text{occ}} = \begin{cases} \varepsilon_{\text{off}} & \text{occupied-sleep (night-only runtime)} \\ \dfrac{16\,\varepsilon_{\text{pk}} + 8\,\varepsilon_{\text{off}}}{24} & \text{occupied all day} \\ \varepsilon_{\text{pk}} & \text{unoccupied } H \text{ hr},\ H \le 16 \\ \dfrac{16\,\varepsilon_{\text{pk}} + (H-16)\,\varepsilon_{\text{off}}}{H} & \text{unoccupied } H \text{ hr},\ H > 16 \end{cases} \tag{9}$$

where $\varepsilon_{\text{pk}} = 1.041$ and $\varepsilon_{\text{off}} = 0.976$ (in terms of lbs/kWh) are the PJM peak and off-peak marginal factors and H is hours-away per day [37]. Occupied-sleep returns the off-peak factor because the home's conditioning runtime falls entirely in the overnight off-peak window.

Comfort uses a tent function anchored to ASHRAE 55 PMV-optimal setpoints [82, 83, 84]:

$$C_{\text{comfort}} = \max\bigl(0,\ 10 - |T_{\text{in}} - T^*|\bigr) \tag{10}$$

where C_{comfort} is a 0–1 comfort score, T_{in} is the indoor setpoint, T^* is the context-dependent optimal setpoint used as the peak of the tent function (76 °F in cooling mode, 70 °F in heating mode, consistent with ASHRAE 55-2020 sedentary PMV bands [82]). A small penalty $(n-3) \times 0.3 \times |T_{\rm in} - T^*|/3.0$ scales with household size beyond three occupants. Practicality starts at 10 and subtracts an asymmetric extremity penalty for setpoints beyond adoption-comfort bounds:

$$\delta(T_{\rm in}) = \begin{cases} 1.0\,(71 - T_{\rm in}) & \text{cooling, } T_{\rm in} < 71 \\ 1.5\,(T_{\rm in} - 82) & \text{cooling, } T_{\rm in} > 82 \\ 1.8\,(63 - T_{\rm in}) & \text{heating, } T_{\rm in} < 63 \\ 0.8\,(T_{\rm in} - 76) & \text{heating, } T_{\rm in} > 76 \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

where $\delta(T_{\rm in})$ is the extremity penalty (score points per °F beyond adoption-comfort bounds), with the steepest slopes on the least-tolerable directions. The pre-feasibility practicality score is $P_{\rm prac} = \mathrm{clip}_{[1.5,\,10]}\big(10 - \delta(T_{\rm in})\big)$.

The result is then multiplied by an outdoor–indoor ΔT feasibility factor (1.0, 0.95, 0.85, 0.70 for $\Delta T < 10$, < 20, < 35, ≥ 35 °F), signaling operation near system limits. Finally, age- and maintenance-driven efficiency degradation enters here as a reliability factor: an older or poorly maintained unit is a less practical choice to rely on, so the score is scaled by $(1 - D)$ where

$$D = \min\Big(0.30,\ \delta_{\rm maint}\big[\,1.5\min(a, 10) + 0.5\max(a - 10, 0)\,\big]\Big) \tag{12}$$

is the fraction of efficiency lost (capped at 30%), a is HVAC age in years, and $\delta_{\rm maint}$ is the base annual loss rate (0.005, 0.010, 0.015 per year for good, moderate, and poor upkeep); the first ten years degrade at $1.5\,\delta_{\rm maint}$ and later years at $0.5\,\delta_{\rm maint}$ [85, 86, 87]. The final practicality score is clipped to $[1.5, 10]$.

An "Off" alternative (no active conditioning) is scored with zero decision-period energy, cost, and emissions; its comfort and practicality are instead evaluated against the free-float indoor temperature the building drifts to with the system off — outdoor $+5$ °F in cooling season and $+10$ °F in heating season, reflecting solar and internal gains that hold a closed, occupied house above outdoor air temperature [58, 82, 88, 89].

3.6.2. Appliance Calculator

Energy cost per cycle is

$$C = E_{\rm cyc} \cdot r(t, \ell) \tag{13}$$

where $E_{\rm cyc}$ is kWh/cycle and $r(t, \ell)$ is the location- and time-specific TOU rate resolved to one of six Pennsylvania utilities (PECO, PPL, West Penn, Penelec, MetEd, Duquesne) [38] (see Appendix F for the full rate structure). . C is per-cycle energy cost (dollars/cycle), $E_{\rm cyc}$ is per-cycle energy use (kWh/cycle), t is the appliance run time (time-of-day), and ℓ is the household location used to resolve the serving utility.

The PJM emissions window (7am–11pm) applies to environmental impact regardless of the utility's billing window, because emissions follow the regional grid. The run-time rate period is selected by mapping location ℓ to a utility $U(\ell)$ and then applying that utility's TOU peak window otherwise.

$$(h^U_{\rm pk,s}, h^U_{\rm pk,e}) : \quad r(t, \ell) = r^U_{\rm pk} \text{ if } h(t) \in [h^U_{\rm pk,s}, h^U_{\rm pk,e}) \text{ and } r(t, \ell) = r^U_{\rm off} \tag{14}$$

where $U(\ell)$ maps location ℓ to a serving utility, $h^U_{\rm pk,s}$ and $h^U_{\rm pk,e}$ are that utility's TOU peak-window start and end hours, $r^U_{\rm pk}$ and $r^U_{\rm off}$ are the utility's peak and off-peak energy prices ($/kWh), and $h(t)$ maps time t to an hour-of-day on a 24-hour clock. In the implementation, the six utilities use distinct peak windows (e.g., 2–6pm for PECO and PPL, 2–9pm for most FirstEnergy utilities). The appliance calculator does not model weekend or holiday peak-hour pricing; all days use weekday TOU rates. Seasonal peak-window shifts are also unmodeled: PPL's winter (Dec–May) peak window is 4–8pm versus the 2–6pm summer window applied here; the summer window is used year-round because scenarios carry no season or date field. This simplification shifts the optimal scheduling window by 1–2 hours for some utilities in winter months. Correcting it would require a date-aware scenario generator and re-running all appliance scenarios; the current implementation trades this precision for consistency with the HVAC and Shower calculators, which also use fixed seasonal assumptions.

Comfort is a piecewise decay from 10 at zero delay, with appliance-specific tolerance ceilings of 12 hr (dishwashers), 8 hr (washers), and 6 hr (dryers) [59, 60], plus an additive noise penalty scaled by housing type for runs after 10 pm [90, 91, 92, 93, 94] and a household-size penalty. Apartment and Condo units receive the largest noise multiplier ($\kappa = 1.5$) because shared walls transmit vibration directly to adjacent units; Townhouse and Rowhouse receive an intermediate multiplier ($\kappa = 1.2$) reflecting one or two party walls with some buffering; Single-family homes receive the smallest ($\kappa = 0.8$) because the structure is isolated. The household-size penalty $\sigma(n)$ steps from 0.0 for fewer than 3 occupants, to 0.8 for 3–4 occupants, to 1.5 for 5 or more, reflecting the empirical finding that multi-person households experience disproportionate annoyance from delayed appliance cycles [59, 60]. Delays are computed as the minimum circular distance from the baseline time t_0 on a 24-hour clock (avoiding wrap-around artifacts), i.e.,

$$d(t, t_0) = \min\big((h(t) - h(t_0)) \bmod 24,\ (h(t_0) - h(t)) \bmod 24\big), \tag{15}$$

where $d(t, t_0)$ is delay in hours between run time t and baseline time t_0, $h(\cdot)$ maps a time to hour-of-day, and $\bmod 24$ indicates wrap-around on a 24-hour clock. The base comfort component uses the following piecewise schedule on delay hours d:

$$C_{\text{base}}(d) = \begin{cases} 10 & d = 0 \\ 10 - \frac{2}{3}\,d & 0 < d \leq 3 \\ 8 - \frac{1}{2}\,(d-3) & 3 < d \leq 7 \\ 6 - \frac{2}{5}\,(d-7) & 7 < d \leq 12 \\ 2 & d > 12 \end{cases} \tag{16}$$

where $C_{\text{base}}(d)$ is the pre-penalty comfort score (0–1) as a function of delay d. The late-night noise penalty is applied when running between 10pm and 7am and the appliance exceeds a dBA threshold:

$$N(t, \text{type}, \text{housing}) = \begin{cases} 0 & \text{if } h(t) \in [7, 22) \text{ or } \text{dBA}(\text{type}) \leq \theta \\ 2 \cdot \kappa(\text{housing}) & \text{otherwise} \end{cases} \tag{17}$$

where $N(t, \text{type}, \text{housing})$ is a nonnegative noise-disruption penalty in comfort-score points, $\text{dBA}(\text{type})$ is the appliance noise level for the appliance type, $\theta = 45\,\text{dBA}$ is the dBA threshold, and $\kappa(\text{housing})$ is a unitless housing-type multiplier (larger for shared-wall housing). The household-size penalty scales with delay relative to the appliance-specific maximum acceptable delay $d_{\max}(\text{type})$:

$$S(d, n, \text{type}) = \sigma(n) \cdot \min\left(\frac{d}{d_{\max}(\text{type})}, 1\right), \tag{18}$$

where $S(d, n, \text{type})$ is a nonnegative household-size penalty in comfort-score points, n is occupants, $d_{\max}(\text{type})$ is the maximum acceptable delay (hours) for the appliance type, and $\sigma(n)$ is an occupants-based factor (higher for larger households). Final comfort is then

$$C_{\text{comfort}} = \text{clip}_{[0,10]}\big(C_{\text{base}} - N - S\big) \tag{19}$$

where $\text{clip}_{[a,b]}(x) = \min(b, \max(a, x))$ clamps a value to the interval $[a, b]$. Practicality mirrors the delay schedule with additional penalties for late-night timing complexity and household coordination difficulty. The base practicality component is:

$$P_{\text{base}}(d) = \begin{cases} 10 & d = 0 \\ 10 - d & 0 < d \leq 2 \\ 8 - \frac{3}{4}\,(d-2) & 2 < d \leq 4 \\ 6.5 - \frac{1}{2}\,(d-4) & 4 < d \leq 8 \\ 4.5 - \frac{3}{8}\,(d-8) & 8 < d \leq 12 \\ 1.5 & d > 12 \end{cases} \tag{20}$$

where $P_{\text{base}}(d)$ is the pre-penalty practicality score (0–1) as a function of delay d. Timing complexity applies an additive penalty by run-time hour:

$$T(t) = \begin{cases} 2 & 0 \leq h(t) < 6 \\ 1 & 22 \leq h(t) < 24 \\ 0 & \text{otherwise} \end{cases} \tag{21}$$

where $T(t)$ is a nonnegative timing-complexity penalty in practicality-score points and $h(t)$ is hour-of-day. Household coordination difficulty uses the same delay scaling form as in comfort:

$$K(d, n, \text{type}) = \kappa_n(n) \cdot \min\left(\frac{d}{d_{\max}(\text{type})}, 1\right), \tag{22}$$

where $K(d, n, \text{type})$ is a nonnegative household-coordination penalty in practicality-score points and $\kappa_n(n)$ is an occupants-based factor increasing with occupants. Final practicality is $C_{\text{prac}} = \text{clip}_{[1.5,10]}\big(P_{\text{base}} - T - K\big)$, with a daytime floor applied for $h(t) \in [7, 22)$ and $d \leq d_{\max}(\text{type})$. For budget utilization in (5), the appliance calculator converts per-cycle cost to a monthly estimate as:

$$C_{\text{monthly}} = c(\text{type})\, C_{\text{cycle}}, \qquad c(\text{type}) = \begin{cases} 18 & \text{dishwasher} \\ 25 & \text{washer} \\ 24 & \text{dryer} \end{cases} \tag{23}$$

where C_{cycle} is per-cycle energy cost and $c(\text{type})$ is the appliance-specific cycles-per-month estimate, from 215, 300, and 283 cycles/year for dishwashers, washers, and dryers respectively [95, 96, 97].

3.6.3. Shower Calculator.

Mains inlet temperature is interpolated from outdoor temperature between 45 °F (winter, $T_{\text{out}} \leq 32$ °F) and 65 °F (summer, $T_{\text{out}} \geq 75$ °F) [73, 98]. Only the fraction of flow requiring heating reaches the heater; that mixing fraction is:

$$f_{\text{hot}} = \frac{T_{\text{target}} - T_{\text{inlet}}}{T_{\text{heater}} - T_{\text{inlet}}} \tag{24}$$

where f_{hot} is the fraction of total flow drawn as hot water, $T_{\text{target}} = 105$ °F is the standard comfortable delivery temperature [99], T_{inlet} is mains inlet temperature, and T_{heater} is the water-heater setpoint temperature. Shower energy is then:

$$E_{\text{kWh}} = \frac{\dot{V} \cdot f_{\text{hot}} \cdot 8.33 \cdot (T_{\text{heater}} - T_{\text{inlet}}) \cdot \tau}{3412 \cdot \eta} \tag{25}$$

where E_{kWh} is shower energy use (kWh), $\dot{V}$ is total flow rate (GPM), τ is shower duration (min), $\eta = 0.92$ is the unified energy factor for a 40–55 gal electric tank [100], and 3412 BTU/kWh is the standard conversion; the constant 8.33 lb/gal converts gallons to pounds of water.

Environmental impact is water volume:

$$V = \dot{V} \cdot \tau \tag{26}$$

measured in gallons, where $\dot{V}$ is total flow rate (GPM) and τ is shower duration (min).

Energy cost uses a flat electricity rate p_e:

$$C_{\$} = p_e\, E_{\text{kWh}}. \tag{27}$$

where $C_{\$}$ is shower energy cost (dollars) and p_e is the electricity price ($/kWh). Comfort is a piecewise function peaking at the REU2016 average of 7.8 min [52, 53], with additive penalties for temperature adequacy (CDC Legionella thresholds [61]) and household contention. The implementation uses a duration-based piecewise base comfort $C_{\text{base}}(\tau)$ (with a short-duration cubic ramp to reflect severely rushed showers), then subtracts a temperature penalty $\Delta_{\text{temp}}(T_{\text{heater}})$ and a household contention penalty $\Delta_{\text{cont}}(\tau, n)$, and clips to $[0, 10]$. The comfort-optimal duration is the 7.8 min REU2016 mean at moderate outdoor temperatures but is shifted upward in cold weather to at most

10.3 min, the observed winter field-mean envelope. This cold-weather elasticity is an explicit modeling assumption supported only in direction by the literature [101, 102, 52, 53], and warm-weather scoring is unchanged:

$$C_{\text{comfort}}(\tau, T_{\text{heater}}, n) = \text{clip}_{[0,10]}\Big(C_{\text{base}}(\tau) - \Delta_{\text{temp}}(T_{\text{heater}}) - \Delta_{\text{cont}}(\tau, n)\Big). \tag{28}$$

where τ is shower duration (min), n is occupants, $C_{\text{base}}(\tau)$ is the base comfort curve, $\Delta_{\text{temp}}(T_{\text{heater}})$ is a temperature-adequacy penalty, and $\Delta_{\text{cont}}(\tau, n)$ is a household-contention penalty.

Practicality additionally penalizes alternatives that exceed available tank capacity. Unlike comfort, the base practicality curve is not centered on the 7.8 min comfort optimum: it rises with duration up to a peak near 12 min before declining, because practicality here measures likelihood of adoption for everyone—how little behavior change an alternative demands of a household that already showers at a given length—rather than momentary preference. A very short shower scores low on practicality (it requires sustained restraint), while a moderately long one is easy to sustain. Pressure toward shorter showers in the MAVT aggregate therefore comes from the cost and environmental criteria and from the comfort optimum, not from practicality.

On the other hand, the capacity constraint below is what makes an over-long shower infeasible for a constrained household. The practicality model uses a duration-based base practicality $P_{\text{base}}(\tau)$ and a capacity constraint computed from hot-water demand. Hot-water per shower is $\tau\,\dot{V}\,f_{\text{hot}}$ (gal), total household hot-water need is $n\,\tau\,\dot{V}\,f_{\text{hot}}$, and available draw is approximated as 0.8 of nominal tank size S:

$$H_{\text{need}} = n\,\tau\,\dot{V}\,f_{\text{hot}}, \qquad H_{\text{avail}} = 0.8\,S. \tag{29}$$

where H_{need} is total hot-water volume required for the household's showers over the relevant period, H_{avail} is available hot-water draw volume, and S is nominal tank capacity (all in gallons). A fixed penalty is applied when $H_{\text{need}} > H_{\text{avail}}$, yielding a clipped final practicality score. For budget utilization in the budget penalty equation, shower monthly cost is estimated from per-shower cost using an average showers-per-person-per-day rate:

$$C_{\text{monthly}} = C_{\text{shower}} \cdot \big(n \cdot 0.9 \cdot 30\big). \tag{30}$$

where C_{shower} is per-shower energy cost, n is occupants, 0.9 is showers/person/day, and 30 is days/month.

3.7. Architecture Implementations

All three architectures call the same underlying model via OpenRouter at temperature 0.3, a moderate value that balances instruction-following fidelity against output diversity [103], with model held constant within each benchmark run so that architecture differences rather than model differences drive the comparison. The study evaluates four models spanning a range of capability, reasoning support, and cost: GPT-OSS-20B (reasoning, low effort), Qwen 3.5 9B (non-reasoning), DeepSeek V4 Flash (hybrid reasoning), and Gemini 3.5 Flash (reasoning, minimal effort). Each architecture is described below in Figure 4.

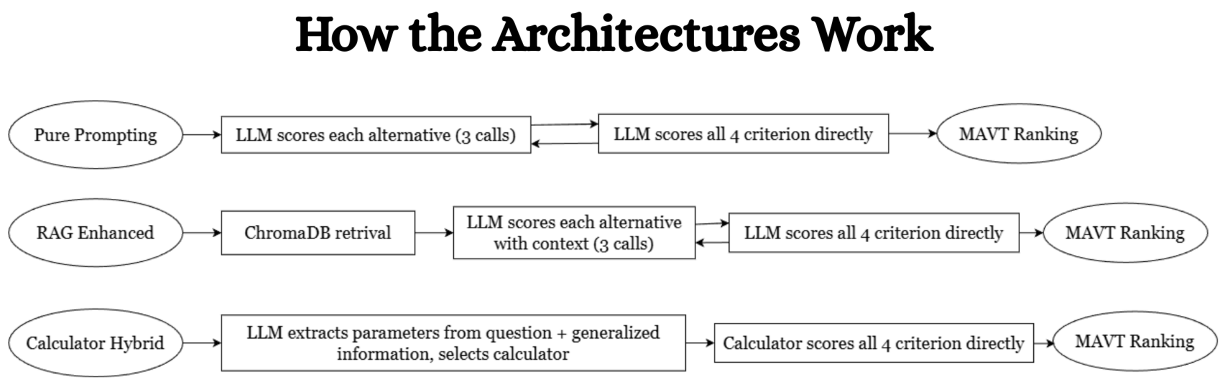

Figure 4: Processing pipeline for each architecture.

3.7.1. $\mathcal{A}_D$ (Direct Prompting)

The $\mathcal{A}_\text{D}$ architecture issues one API call per alternative (three calls per scenario) using a calibrated system prompt that combines role-prompting [104], structured-output constraints [105], and explicit bias-mitigation language drawn from the NIST framework for identifying and managing AI bias [106]. No external context is supplied beyond the scenario description. The system prompt establishes the role and scoring criteria with domain-specific anchors:

```
System prompt:

You are an expert household decision analyst specializing in MCDA. Score alternatives
on four criteria (0-1 scale):

HVAC: energy_cost: good when setpoint demands little system effort; poor when
extreme effort required. environmental: good when system runs efficiently; poor
when high load drives sustained emissions. comfort: good when setpoint feels
comfortable for outdoor temp; poor when noticeably too hot/cold. practicality:
good when easy to maintain without strain; poor when risk of override or system
stress.

Appliance: energy_cost: good during off-peak hours; poor during peak pricing
windows. environmental: good when grid emissions low; poor when peak-period generation
dirtiest. comfort: good with little or no delay; poor when not ready until next
morning. practicality: good when easy to remember and unobtrusive; poor when
noise or coordination difficulty.

Shower: energy_cost: good when short; worsens as duration increases. environmental:
good when water volume low; worsens as duration increases. comfort: good near
typical length; poor when too short or wastefully long. practicality: good when
sustainable daily; poor when too short to maintain or causes hot water contention.

Return ONLY: {"energy_cost": X, "environmental": X, "comfort": X, "practicality":
X} (0.0-1.0). Distinguish between criteria.
```

The user prompt provides the scenario context and asks the model to score one alternative at a time, including the other alternatives as reference so that it doesn't assign the same scores to all alternatives.

3.7.2. $\mathcal{A}_E$ (Example-Guided LLM Scoring)

The $\mathcal{A}_\text{E}$ architecture retrieves the top $k = 3$ most similar scenarios from a ChromaDB vector store before scoring, using the sentence-transformers/all-MiniLM-L6-v2 embedding model. Each retrieved exemplar is rendered as a full worked example: its homeowner-facing parameters, its engineering parameters (R-value, SEER, GPM, kWh/cycle, tank size, heater setpoint, etc.), and each of its alternatives with the four ground-truth criterion scores plus the MAVT aggregate and rank. The exemplar's engineering values are shown so the model can see how those values map to scores. Thus, it can look at overall patterns for similar scenarios to understand how its own scenario may perform. The model is instructed to use the exemplars as reference but to score the target scenario independently. Like $\mathcal{A}_\text{D}$, $\mathcal{A}_\text{E}$ uses three API calls per scenario (one per alternative). The retrieval index is built once from the 90 RAG-only scenarios described in Section 2.1 and is never queried with test scenarios [21].

The system prompt is identical to $\mathcal{A}_\text{D}$ but omits per-criterion anchor descriptions, since the RAG exemplars supply implicit scoring guidance:

```
System prompt (shortened--no per-criterion anchors):

You are an expert household decision analyst specializing in MCDA. Score alternatives
on four criteria (0-1 scale): 1. Energy Cost: Lower costs = higher score. 2.
Environmental Impact: Lower emissions = higher score. 3. Comfort: Higher user
comfort = higher score. 4. Practicality: Easier to implement/maintain = higher
score.

Return ONLY: {"energy_cost": X, "environmental": X, "comfort": X, "practicality":
X} (0.0-1.0). Distinguish between criteria.
```

The user prompt presents the target scenario, the alternative to score, and three retrieved exemplars—each with homeowner-facing parameters, engineering values, and the four ground-truth scores plus MAVT aggregate and rank—instructing the model to "consider how this specific alternative performs given the scenario context."

3.7.3. $\mathcal{A}_H$ (LLM-Parameterized Reference Scoring)

The $\mathcal{A}_{\mathrm{H}}$ architecture issues a single API call per scenario for structured parameter extraction. Known homeowner-visible scenario fields (outdoor temperature, square footage, insulation tier, household size, housing type, utility budget, and location) are passed directly to the calculator without LLM involvement, and the three alternatives are taken verbatim from the scenario sheet—the deterministic calculator, not the LLM, parses the numeric content of each alternative string (setpoint temperature, run time, or shower duration). The LLM is asked only to estimate the genuinely withheld engineering parameters that are absent from the homeowner-visible description: for HVAC, the wall R-value, SEER rating, HVAC age, and occupancy context; for Appliance, the appliance type, kWh/cycle, and current baseline time; for Shower, the numeric GPM (estimated from the homeowner-facing flow-rate label), tank size, and water-heater setpoint. The extracted JSON is validated—each numeric parameter must parse as a finite value within its physically admissible range, and the extracted decision type must match the scenario's known type, or the scenario is recorded as an extraction failure—and then merged with the known fields and passed to the corresponding ground-truth calculator.

The extraction prompt uses decomposed prompting [107] to separate natural-language understanding from quantitative computation. Unlike $\mathcal{A}_{\mathrm{D}}$ and $\mathcal{A}_{\mathrm{E}}$, it does not ask the LLM to score; it asks the LLM to extract engineering parameters:

Extraction prompt:

```
You are a household-systems engineer. Extrapolate missing engineering parameters
from the scenario context.

SCENARIO: {scenario_text}
QUESTION: {question}

INSTRUCTIONS: Identify the Decision Type. Estimate ONLY the parameters below.
Return ONLY valid JSON.

HVAC: {"decision_type": "HVAC", "parameters": {"r_value": <number>, "seer":
<number>, "hvac_age": <number>, "occupancy_context": "occupied_all_day | unoccupied_<hours>
| occupied_sleep"}}

Appliance: {"decision_type": "Appliance", "parameters": {"appliance": "Dishwasher
| Washer | Dryer", "kwh_per_cycle": <number>, "baseline_time": "<e.g. 7pm,
8am>"}}

Shower: {"decision_type": "Shower", "parameters": {"gpm": <number>, "tank_size":
<number>, "water_heater_temp": <number>}}

Return ONLY the JSON, no explanation.
```

Because the alternatives and all homeowner facts are supplied deterministically and only the scenario-level engineering parameters are model-estimated, the calculator does the physics identically to the ground truth. $\mathcal{A}_{\mathrm{H}}$ uses 1 API call per scenario versus 3 for $\mathcal{A}_{\mathrm{D}}$ and $\mathcal{A}_{\mathrm{E}}$. Because the parameters the LLM estimates are scenario-level and identical across the three alternatives, most estimation errors shift all three raw values together and largely cancel in ranking; ranking accuracy is therefore expected to be robust to weak extraction, whereas the score-level error (MAE/RMSE) and the per-alternative appliance baseline time are the measures that will actually reveal the quality of the extraction.

3.8. RAG Ablation Design

Ablation experiments use a stratified 35-scenario subset drawn from the RAG corpus to ensure adequate statistical power. Seven configurations modify one design variable at a time against a $k = 3$ control: retrieval k (1, 5), embedding model (paraphrase-MiniLM-L3-v2 replacing all-MiniLM-L6-v2), exemplar content (hidden engineering parameters included/excluded), score anchoring (scores and ranks present/absent), random exemplar selection (replacing similarity-ranked exemplars with random corpus draws), and a nearest-neighbor offline baseline that assigns retrieved scenario scores directly without LLM rescoring. Each configuration is evaluated across all four models. The $k = 1$ variant tests whether a single high-similarity exemplar provides sufficient guidance; $k = 5$ tests whether broader context improves or dilutes the signal. The embedding-swap tests a different matching architecture. The exemplar-content variants isolate whether the model benefits from seeing the engineering parameters (R-value, SEER, GPM, kWh/cycle) or only the homeowner-facing labels. The score-anchoring variants test whether the model uses the ground-truth scores as calibration anchors or merely as noise. The random-exemplar variant controls for retrieval quality: if retrieval similarity matters, random exemplars should perform worse than similarity-ranked ones. The nearest-neighbor offline baseline tests whether unmodified exemplar copying provides ranking signal.

3.9. Evaluation Metrics

Accuracy is evaluated on two axes. Score-level error uses Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) between architecture scores and ground-truth scores on the 0–1 MAVT scale, computed separately for each criterion and aggregated across criteria. Ranking accuracy uses Kendall's τ between the architecture's induced ranking of the three alternatives and the ground-truth ranking, together with Top-1 accuracy (architecture's best-ranked alternative matches ground truth) and Top-2 accuracy (ground-truth top alternative appears in the architecture's top two). For each scenario, Kendall's τ is computed between the ground-truth rank and the architecture's rank over the three alternatives, then averaged across all 195 scenarios by arithmetic mean. With $n = 3$ alternatives, per-scenario τ can take only discrete values $(-1, -\frac{1}{3}, \frac{1}{3}, 1)$ before averaging. Efficiency is evaluated using total tokens (input plus output) and cost, both summed across the 195-scenario test set.

Failure is recorded per scenario via a reserved sentinel value (1928) that marks any of the failure modes listed in Table 6. Accuracy metrics are conditioned on success: scenarios containing any sentinel are excluded from MAE/RMSE and ranking computations and counted separately in the failure rate. $\mathcal{A}_{\mathrm{D}}$ and $\mathcal{A}_{\mathrm{E}}$ issue one call per alternative and mark the whole scenario failed if any alternative fails; $\mathcal{A}_{\mathrm{H}}$ issues a single extraction call per scenario, so its failure unit is the scenario.

Table 6: Failure modes and their trigger conditions.

Failure Mode	Trigger	Applies To
Malformed output	Non-JSON or unparseable response	All architectures
Missing score	Criterion score absent from JSON	$\mathcal{A}_{\mathrm{D}}$, $\mathcal{A}_{\mathrm{E}}$
Out-of-bounds score	Score < 0 or > 10	All architectures
Invalid score type	Non-numeric score value	$\mathcal{A}_{\mathrm{D}}$, $\mathcal{A}_{\mathrm{E}}$
Failed extraction	Non-JSON, invalid decision type, or non-finite parameter	$\mathcal{A}_{\mathrm{H}}$ only
Calculation error	Ground-truth calculator exception	$\mathcal{A}_{\mathrm{H}}$ only

4. Results

4.1. Overall Ranking Accuracy

Table 7 shows each model's performance across architectures (full per-model results by decision type appear in Appendix C). $\mathcal{A}_{\mathrm{H}}$ dominates on every metric for all four models: τ ranges from 0.877 (Qwen) to 0.925 (Gemini), Top-1 from 90.3% (Qwen) to 93.3% (Gemini). $\mathcal{A}_{\mathrm{E}}$ places second across all models: τ from 0.282 (Qwen) to 0.378 (Gemini), Top-1 from 49.7% (DeepSeek) to 54.4% (Gemini). $\mathcal{A}_{\mathrm{D}}$ ranks third: τ from -0.002 (Qwen) to 0.169 (Gemini), Top-1 from 31.8% (Qwen) to 40.0% (DeepSeek), near the random baseline of $\tau = 0$ for a three-alternative ranking. The most interesting finding is while the model success order stays almost exactly the same, for $\mathcal{A}_{\mathrm{H}}$, the gap is much smaller than in $\mathcal{A}_{\mathrm{D}}$

The ranking metrics are partly structurally favored in $\mathcal{A}_{\mathrm{H}}$ (see Section 3.7 for the structural argument); the score-level error (MAE/RMSE) is the stronger evidence of extraction fidelity.

The ordering holds across all nine weight perturbations (Section 3.3). Pairwise differences between architectures are significant under Wilcoxon signed-rank testing ($p < 0.001$ for each adjacent pair), applied to per-scenario Kendall's τ within each model and combined across models using Stouffer's method (see Appendix I for full results).

ICC(2,1) for MAE is 0.80–0.91 across architectures: model choice drives most score-level error. ICC for Top-1 accuracy is 0.08–0.27: scenario difficulty drives correct-alternative identification. Cliff's δ between $\mathcal{A}_{\mathrm{E}}$ and $\mathcal{A}_{\mathrm{H}}$ is 0.66–0.80 for MAE and 0.55–0.77 for τ (both large effects). Bootstrap 95% BCa intervals for $\mathcal{A}_{\mathrm{H}}$ tau [0.892, 0.907] and MAE [0.050, 0.060] do not overlap $\mathcal{A}_{\mathrm{E}}$ or $\mathcal{A}_{\mathrm{D}}$ intervals on any metric.

The MAE and RMSE values quantify how closely each architecture's scores match the ground truth. $\mathcal{A}_{\mathrm{H}}$ achieves overall MAE of 0.051 on the 0–1 MAVT scale, meaning score predictions deviate from the true value by roughly

Metric	Gemini $\mathcal{A}_D$	Gemini $\mathcal{A}_E$	Gemini $\mathcal{A}_H$	DeepSeek $\mathcal{A}_D$	DeepSeek $\mathcal{A}_E$	DeepSeek $\mathcal{A}_H$
Kendall's τ	0.189 ± 0.278	0.405 ± 0.315	$\mathbf{0.920 \pm 0.062}$	0.157 ± 0.259	0.338 ± 0.221	$\mathbf{0.895 \pm 0.071}$
Top-1 (%)	36.9 ± 15.1	56.1 ± 22.9	$\mathbf{92.9 \pm 4.9}$	37.5 ± 16.4	49.9 ± 13.5	$\mathbf{90.5 \pm 6.5}$
MAE	0.218 ± 0.051	0.127 ± 0.029	$\mathbf{0.048 \pm 0.013}$	0.231 ± 0.026	0.139 ± 0.019	$\mathbf{0.046 \pm 0.009}$
RMSE	0.288 ± 0.064	0.194 ± 0.035	$\mathbf{0.097 \pm 0.028}$	0.300 ± 0.033	0.204 ± 0.021	$\mathbf{0.091 \pm 0.015}$
RMSE/MAE	1.32 ± 0.03	1.54 ± 0.07	2.02 ± 0.07	1.30 ± 0.01	1.47 ± 0.06	2.00 ± 0.20

Metric	GPT-OSS $\mathcal{A}_D$	GPT-OSS $\mathcal{A}_E$	GPT-OSS $\mathcal{A}_H$	Qwen $\mathcal{A}_D$	Qwen $\mathcal{A}_E$	Qwen $\mathcal{A}_H$
Kendall's τ	0.041 ± 0.129	0.335 ± 0.198	$\mathbf{0.898 \pm 0.073}$	0.011 ± 0.065	0.280 ± 0.127	$\mathbf{0.880 \pm 0.029}$
Top-1 (%)	33.2 ± 6.0	51.9 ± 11.8	$\mathbf{91.6 \pm 6.5}$	30.0 ± 6.3	50.9 ± 7.9	$\mathbf{89.8 \pm 2.8}$
MAE	0.242 ± 0.031	0.145 ± 0.027	$\mathbf{0.052 \pm 0.002}$	0.235 ± 0.031	0.169 ± 0.017	$\mathbf{0.071 \pm 0.030}$
RMSE	0.305 ± 0.035	0.212 ± 0.034	$\mathbf{0.101 \pm 0.006}$	0.294 ± 0.033	0.236 ± 0.019	$\mathbf{0.152 \pm 0.056}$
RMSE/MAE	1.27 ± 0.03	1.47 ± 0.04	1.95 ± 0.05	1.25 ± 0.03	1.40 ± 0.04	2.18 ± 0.15

Table 7: Overall performance by model (5-run mean ± std, 195 scenarios). Best architecture per metric per model is bolded.

5% on average. Its RMSE of 0.105 (Table 7) yields an RMSE/MAE ratio of 2.06—the highest among the three architectures—indicating that while average error is low, a small number of scenarios (primarily shower cases requiring GPM estimation) produce larger deviations that pull RMSE above twice the MAE. By contrast, $\mathcal{A}_D$ MAE of 0.219 means individual scores routinely miss by more than 0.2 points on a 1-point scale, and $\mathcal{A}_E$ at 0.138 falls in between—usable for rough ranking but not for reliable score comparison.

The ranking metrics translate these score errors into decision quality. Kendall's $\tau > 0.9$, as achieved by $\mathcal{A}_H$ across all four models, means the architecture identifies the correct ranking order in nearly every scenario—comparable to the agreement two human experts would reach on the same decision. τ between 0.3 and 0.5, as in $\mathcal{A}_E$, provides a weak but non-random signal: the tool is better than chance but unreliable for any single decision. τ near zero, as in $\mathcal{A}_D$, offers no actionable ranking information. At the Top-1 level, 91.7% accuracy ($\mathcal{A}_H$) means the recommended alternative is genuinely the best option nine times out of ten; 35.4% ($\mathcal{A}_D$) is worse than a coin flip for two alternatives.

4.2. Baseline Comparison

Table 8 extends the comparison beyond the three LLM architectures to two non-LLM baselines, all evaluated on the same 195-scenario test set. Metrics are reported as per-type arithmetic means (averaging HVAC, Appliance, and Shower results equally) so that each decision type contributes equally regardless of scenario count; this prevents FixedDefault's strong HVAC performance from inflating its aggregate score. The Fixed-Default baseline substitutes fixed engineering defaults—68°F setpoint for HVAC, 7PM schedule for appliances, 8-minute shower duration—into the same physics calculators used by the ground truth. It runs the same physics but with a hardcoded "typical" choice. FixedDefault achieves high HVAC accuracy (the 68°F setpoint is near-optimal for most Pennsylvania scenarios) but collapses on Appliance ($\tau = 0.097$) and Shower ($\tau = 0.800$), yielding a per-type average of only $\tau = 0.610$. NearestNeighbor ($k = 3$ majority vote over the RAG corpus) averages $\tau = 0.386$ across types, placing it between $\mathcal{A}_D$ and $\mathcal{A}_E$ and demonstrating that unmodified exemplar copying provides partial but limited ranking signal.

Table 8: Incremental contribution of each system over Fixed-Default baseline (per-type arithmetic means of Top-1 accuracy and Kendall's τ).

System	Top-1	Δ vs FixedDef	Kendall τ	Δ vs FixedDef
FixedDefault	0.7254	0.0000	0.6102	0.0000
NearestNeighbor	0.5775	−0.1479	0.3863	−0.2239
Direct_LLM_Scoring	0.3583	−0.3671	0.1040	−0.5062
Example-Guided_LLM_Scoring	0.5260	−0.1994	0.3527	−0.2576
LLM-Parameterized_Reference_Scoring	0.9160	+0.1906	0.8990	+0.2888

Per the structural argument in Section 3.7, the score-level error is the stronger evidence of extraction fidelity, shown per model in Table 7. Across the four models, $\mathcal{A}_{\mathrm{H}}$ MAE ranges from 0.043 (DeepSeek) to 0.068 (Qwen), representing reductions of 63–81% over $\mathcal{A}_{\mathrm{D}}$ (0.208–0.230) and 47–73% over $\mathcal{A}_{\mathrm{E}}$ (0.127–0.159) per model. RMSE follows the same ranking across all four models.

4.3. Criterion-Level Error

Table 9 and Figure 5 break MAE down by criterion and model. $\mathcal{A}_{\mathrm{H}}$ achieves near-zero error on Comfort and Practicality across all four models: Comfort MAE ranges from 0.008 (Qwen) to 0.013 (DeepSeek), Practicality from 0.016 (Qwen) to 0.022 (GPT-OSS). This is structural: the calculator handles both criteria deterministically, with the LLM only estimating scenario-level parameters (occupancy, tank size) that shift all three alternatives together. This confirms the central-tendency prediction made in Section 3.3.

Error for all three architectures concentrates on Energy Cost and Environmental Impact. $\mathcal{A}_{\mathrm{D}}$ Environmental MAE is 0.247 (Gemini) and 0.263 (Qwen); Energy Cost is 0.247 and 0.243. $\mathcal{A}_{\mathrm{E}}$ reduces Environmental to 0.132 (Gemini) and 0.169 (Qwen), Energy Cost to 0.143 and 0.171. $\mathcal{A}_{\mathrm{H}}$ cuts further: Environmental 0.091 (Gemini) and 0.141 (Qwen), Energy Cost 0.069 and 0.105; Qwen carries the largest residual on both, indicating its smaller parameter count produces less precise domain estimates. The Environmental–Energy Cost gap persists across models: $\mathcal{A}_{\mathrm{D}}$ Environmental MAE exceeds Energy Cost for both, indicating LLMs struggle more with emissions estimation than with dollar costs.

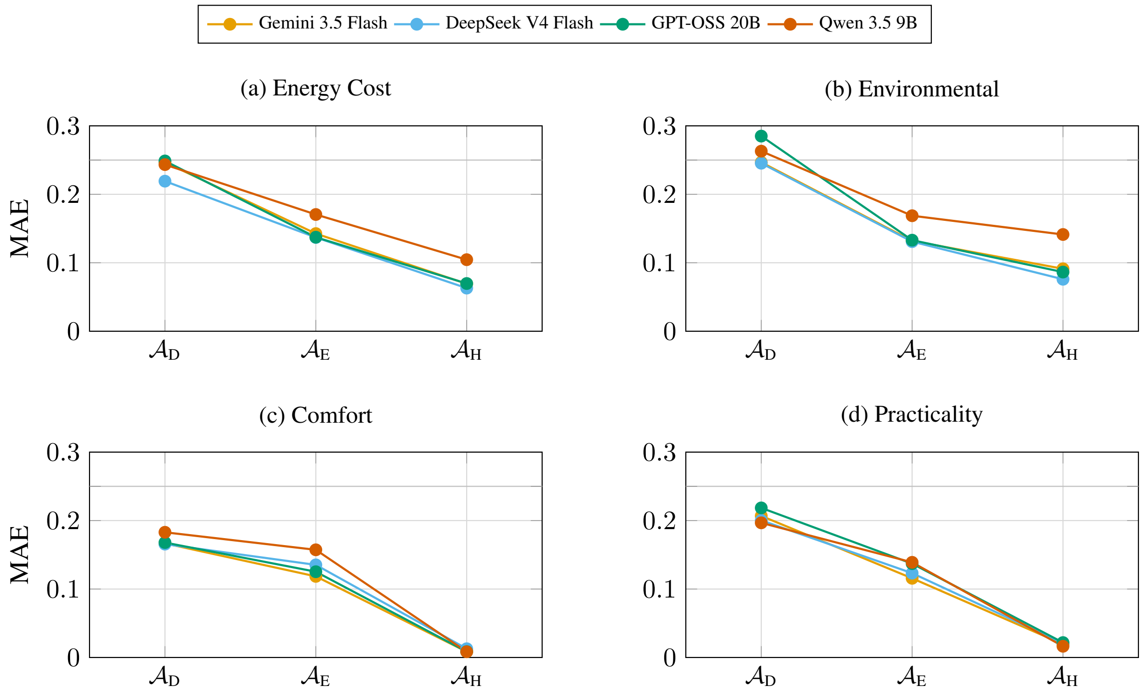

Figure 5: MAE by criterion and model (5-run mean, 195 scenarios each). Each line is one model; the x-axis is architecture ($\mathcal{A}_{\mathrm{D}} \rightarrow \mathcal{A}_{\mathrm{E}} \rightarrow \mathcal{A}_{\mathrm{H}}$).

Criterion	Gemini (strongest) $\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$	Qwen (weakest) $\mathcal{A}_D$	$\mathcal{A}_E$	$\mathcal{A}_H$
Energy Cost	0.247	0.143	**0.069**	0.243	0.171	**0.105**
Environmental	0.247	0.132	**0.091**	0.263	0.169	**0.141**
Comfort	0.167	0.119	**0.009**	0.183	0.157	**0.008**
Practicality	0.207	0.116	**0.020**	0.197	0.139	**0.016**

Environmental MAE for Shower reflects water-consumption proxy (gallons); see Section 3.6.

Table 9: MAE per criterion by architecture, strongest (Gemini) and weakest (Qwen) model. $\mathcal{A}_H$ bolded per criterion.

4.4. Performance by Decision Type

Table 10 and Figure 6 report τ by decision type, with each line in Figure 6 representing one model. $\mathcal{A}_H$ excels across all three decision types for every model. On HVAC, individual model τ values range from 0.876 (Qwen) to 0.981 (Gemini). On Appliance, three models reach $\tau = 0.959$ and Qwen reaches $\tau = 0.897$. Shower is the weakest type due to GPM estimation limits, with model τ from 0.789 (DeepSeek) to 0.856 (Qwen), still far above the other architectures.

$\mathcal{A}_E$ shows wide model-level variance on HVAC: Gemini ($\tau = -0.076$) and DeepSeek ($\tau = 0.048$) are near-random, while GPT-OSS ($\tau = 0.267$) and Qwen ($\tau = 0.191$) achieve modest positive correlation. On Appliance, Gemini ($\tau = 0.456$) and DeepSeek ($\tau = 0.364$) lead; GPT-OSS ($\tau = 0.067$) and Qwen ($\tau = 0.241$) lag. Shower is the strongest type for $\mathcal{A}_E$ across all models: Gemini ($\tau = 0.822$), GPT-OSS ($\tau = 0.722$), DeepSeek ($\tau = 0.700$), and Qwen ($\tau = 0.433$).

$\mathcal{A}_D$ is near-random or below on HVAC for all four models: Gemini -0.048, DeepSeek -0.124, GPT-OSS 0.238, Qwen 0.057. All four models score negative on Appliance (range -0.190 to -0.026). Shower diverges: Gemini (0.633) and DeepSeek (0.722) show meaningful positive correlation, while GPT-OSS (0.100) and Qwen (0.067) remain near-random. The shower advantage for $\mathcal{A}_D$ concentrates in Gemini and DeepSeek, likely because shower scoring depends on duration and flow rate, which are easier to read from scenario text than HVAC load calculations or appliance TOU scheduling.

Appliance results for $\mathcal{A}_H$ track extraction quality since the baseline time parameter varies across alternatives, making it the strongest test of extraction fidelity.

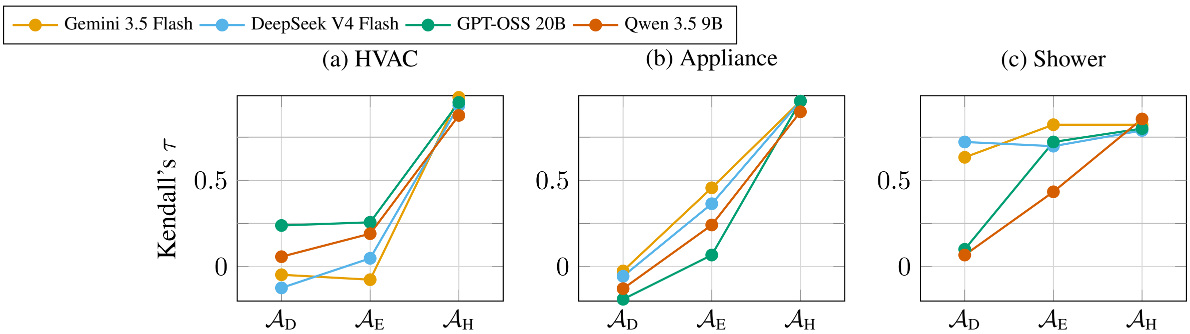

Figure 6: Kendall's τ by decision type and model (5-run mean, 195 scenarios each). Each line is one model; the x-axis is architecture ($\mathcal{A}_D \to \mathcal{A}_E \to \mathcal{A}_H$). The dotted line marks $\tau = 0$ (random ranking).

	$\mathcal{A}_D$		$\mathcal{A}_E$		$\mathcal{A}_H$	
Decision Type	τ	Top-1 (%)	τ	Top-1 (%)	τ	Top-1 (%)
HVAC	0.031 ± 0.136	32.5 ± 12.4	0.107 ± 0.132	33.6 ± 11.1	$\mathbf{0.936 \pm 0.037}$	$\mathbf{92.5 \pm 4.1}$
Appliance	-0.100 ± 0.064	25.8 ± 4.1	0.282 ± 0.146	53.8 ± 11.1	$\mathbf{0.944 \pm 0.027}$	$\mathbf{96.9 \pm 1.9}$
Shower	0.381 ± 0.299	49.2 ± 20.6	0.669 ± 0.144	70.4 ± 7.3	$\mathbf{0.817 \pm 0.025}$	$\mathbf{85.4 \pm 3.8}$
Overall	0.095 ± 0.072	35.4 ± 3.0	0.339 ± 0.035	51.7 ± 1.7	$\mathbf{0.902 \pm 0.017}$	$\mathbf{91.7 \pm 1.1}$

Table 10: Ranking accuracy by decision type (pooled across 4 models, 5 runs each, 195 scenarios). Top-2 accuracy appears in Appendix H.

4.5. *Sensitivity Analysis*

Table 11: The architecture ordering ($\mathcal{A}_H > \mathcal{A}_E > \mathcal{A}_D$) is preserved across all perturbations (Section 4.1).

Scenario	w_{Ene}	w_{Env}	w_{Com}	w_{Pra}
Baseline	0.3000	0.3500	0.2000	0.1500
Ene $+0.05$	**0.3500**	0.3333	0.1833	0.1333
Ene -0.05	**0.2500**	0.3667	0.2167	0.1667
Env $+0.05$	0.2833	**0.4000**	0.1833	0.1333
Env -0.05	0.3167	**0.3000**	0.2167	0.1667
Com $+0.05$	0.2833	0.3333	**0.2500**	0.1333
Com -0.05	0.3167	0.3667	**0.1500**	0.1667
Pra $+0.05$	0.2833	0.3333	0.1833	**0.2000**
Pra -0.05	0.3167	0.3667	0.2167	**0.1000**
Equal	0.2500	0.2500	0.2500	0.2500

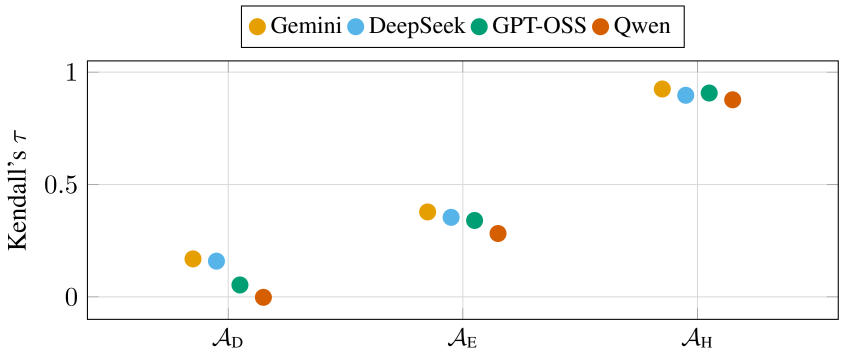

Figure 7: Per-model Kendall's τ under the baseline weight scenario for each architecture. Each dot represents one LLM; color encodes the model. The per-model values confirm the architecture ordering ($\mathcal{A}_H > \mathcal{A}_E > \mathcal{A}_D$) holds for every model individually, with no overlap between architecture groups.

The sensitivity analysis (Table 11) confirms that the architecture ordering is preserved under ± 0.05 perturbations of each weight. Figure 7 visualizes the per-model Kendall's τ under the baseline weights: $\mathcal{A}_H$ achieves $\tau > 0.87$ for every model, $\mathcal{A}_E$ ranges from 0.28 to 0.38, and $\mathcal{A}_D$ falls between -0.002 and 0.17, with no overlap between architecture groups. Full per-model sensitivity results appear in Appendix D.

4.6. *RAG Ablation*

Table 12 summarizes a RAG ablation study conducted on a 35-scenario subset from the RAG corpus with seven configurations across four models, plus an offline nearest-neighbor baseline. The nearest-neighbor offline baseline

Configuration	Overall τ	95% CI	Overall MAE
Retrieval $k = 5$	0.386	$[0.285, 0.486]$	0.079
Retrieval $k = 1$	0.364	$[0.250, 0.474]$	0.093
Alt. embedding (paraphrase-MiniLM-L3-v2)	0.326	$[0.208, 0.443]$	0.081
Control ($k = 3$, all-MiniLM-L6-v2)	0.316	$[0.203, 0.426]$	0.086
Exemplars w/o hidden params	0.300	$[0.181, 0.413]$	0.085
Descriptions w/o scores or ranks	0.109	$[-0.008, 0.219]$	0.142
Random exemplars ($k = 3$)	0.062	$[-0.058, 0.183]$	0.152
Nearest-neighbor (offline)	-0.039	$[-0.333, 0.261]$	0.096

Evaluated on a 35-scenario subset with 4 models. 95% CIs are bias-corrected bootstrap intervals. Nearest-neighbor assigns retrieved scenario scores directly without LLM rescoring.

Table 12: RAG ablation: Kendall's τ and MAE by configuration.

assigns retrieved scenario scores directly without LLM rescoring and yields $\tau = -0.039$ (95% CI: $[-0.333, 0.261]$), below zero and confirming that naive exemplar copying provides no ranking signal.

Retrieval $k = 5$ achieves the highest overall τ (0.386), marginally above $k = 1$ (0.364), with both configurations outperforming the $k = 3$ control (0.316). The exemplars_no_hidden_params configuration ($\tau = 0.300$) performs comparably to the control, indicating that the engineering parameters (R-value, SEER, GPM, kWh/cycle) contribute little additional signal. The alternate embedding model (paraphrase-MiniLM-L3-v2) yields $\tau = 0.326$, comparable to the control. The descriptions_no_scores_ranks configuration, which omits all quantitative scores from the retrieved exemplars, drops to $\tau = 0.109$—the lowest among the LLM-based configurations—confirming that exemplar scores provide critical anchoring information. The random-exemplar variant ($\tau = 0.062$) performs even worse, demonstrating that retrieval similarity matters: semantic matching selects exemplars that are more informative than random corpus draws. The nearest-neighbor offline baseline yields $\tau = -0.039$, confirming that exemplar copying without LLM rescoring provides no useful signal (Figure 8).

A Friedman test across configurations is marginally significant on Kendall's τ ($\chi^2_7 = 12.45$, $p = 0.087$) and highly significant on MAE ($\chi^2_7 = 37.53$, $p < 0.001$). Post-hoc Wilcoxon signed-rank tests with Holm correction confirm that the control, retrieval, exemplar-content, and alternate-embedding configurations form a statistically indistinguishable cluster (pairwise $p_{\text{Holm}} > 0.97$), while random exemplars and descriptions without scores are significantly worse ($p_{\text{Holm}} < 0.003$ versus every configuration in the top cluster). The score MAE tells a consistent story: the top cluster ranges from 0.079 to 0.086, while descriptions without scores (0.142) and random exemplars (0.152) nearly double the error.

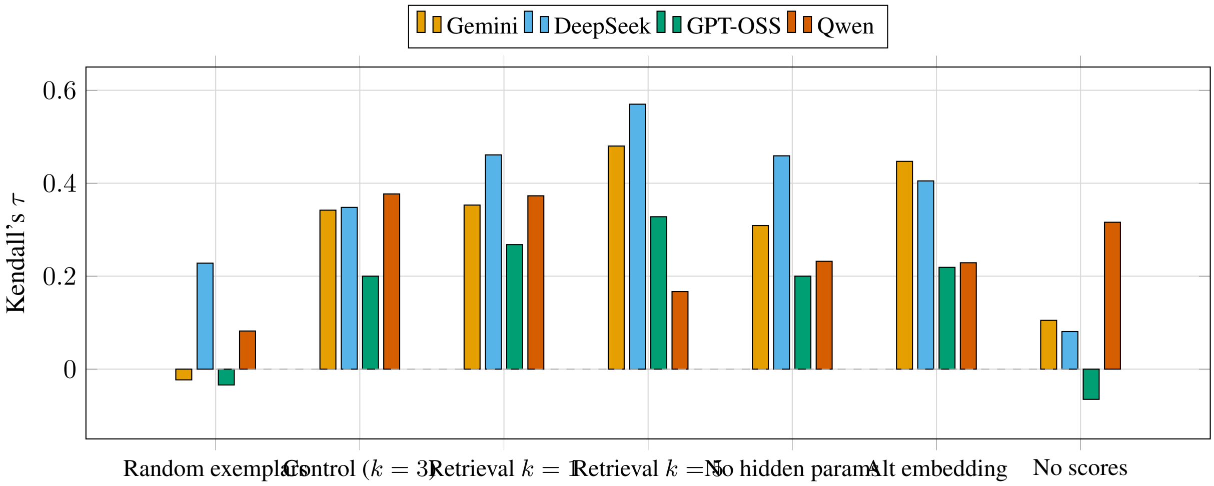

Figure 8: RAG ablation results: Kendall's τ by configuration and model on the 35-scenario subset. Configurations are ordered by overall τ. Dashed line marks $\tau = 0$.

4.7. Efficiency and Cost

Table 13 reports API usage and costs based on OpenRouter pricing as of July 15, 2026. $\mathcal{A}_{\text{H}}$ makes one API call per scenario versus three for $\mathcal{A}_{\text{D}}$ and $\mathcal{A}_{\text{E}}$, and its shorter prompts average 800 input+output tokens per scenario versus $\sim$1,200 for $\mathcal{A}_{\text{D}}$ and $\sim$2,800 for $\mathcal{A}_{\text{E}}$ (which embeds three retrieved exemplars per call). Cost per 195-scenario run ranges from \$0.01 (Qwen, GPT-OSS) to \$0.28 (Gemini). $\mathcal{A}_{\text{D}}$ costs 2–6$\times$ more (\$0.02–\$0.80) and $\mathcal{A}_{\text{E}}$ is the most expensive (\$0.03–\$1.28), driven by exemplar token overhead. Even the priciest configuration stays under \$1.30 per full evaluation. Across four models spanning two orders of magnitude in parameter count and API cost, $\mathcal{A}_{\text{H}}$'s τ varies only between 0.877 and 0.925 (range 0.048), while $\mathcal{A}_{\text{D}}$ spans 0.171 and $\mathcal{A}_{\text{E}}$ spans 0.096—the architecture compresses the capability gap, making cheaper models viable in production with minimal accuracy loss.

Failure rates are near-zero across all architectures and models. The single exception is GPT-OSS $\mathcal{A}_{\text{H}}$, which shows a 12.8% extraction failure rate on parameter parsing—20 to 26 scenarios per run, all HVAC, all triggered by `FAILED_EXTRACTION_INVALID_PARAMETERS` (see Appendix G for a worked example). The five-run averaging protocol recovers most of these: when a scenario fails in one run but succeeds in another, the successful run's score is used. After matching, 0% of scenarios remain unscored across all five runs. This recovery mechanism is fair because the same protocol applies to all architectures; architectures with zero failure rates simply never need it. Without cross-run recovery, GPT-OSS $\mathcal{A}_{\text{H}}$'s pooled τ is 0.886 and Top-1 is 90.8%, versus 0.897 and 91.6% with recovery.

To assess the impact of GPT-OSS extraction failures on per-run metrics, Table 14 reports the best and worst runs for GPT-OSS $\mathcal{A}_{\text{H}}$. The best run (fewest failures) achieves $\tau = 0.906$ and MAE of 0.039; the worst run (most failures) achieves $\tau = 0.886$ and MAE of 0.054. The 0.020 spread in τ and 0.015 spread in MAE are small relative to the between-architecture gaps, confirming that even GPT-OSS's worst run substantially outperforms the best runs of $\mathcal{A}_{\text{E}}$ or $\mathcal{A}_{\text{D}}$.

All results in this paper follow an aggregate-then-evaluate protocol: for each scenario, the five per-run scores are averaged first, producing a single consensus score per criterion per alternative; all metrics (τ, MAE, Top-1 accuracy, etc.) are then computed once on these averaged scores. This protocol is preferred over the compute-per-run-then-average alternative because it directly measures the decision quality of the score a household would actually receive (the five-run consensus), rather than averaging metrics that each reflect a different noise realization. The tradeoff is that it masks run-to-run variance in the final metrics; the boxplots in Figures 9–H.12 and the GPT-OSS per-run analysis above address this directly.

Architecture	Gemini	DeepSeek	GPT-OSS	Qwen
$\mathcal{A}_{\mathrm{D}}$	\$0.80	\$0.09	\$0.02	\$0.04
$\mathcal{A}_{\mathrm{E}}$	\$1.28	\$0.08	\$0.03	\$0.08
$\mathcal{A}_{\mathrm{H}}$	\$0.28	\$0.01	\$0.01	\$0.01

Table 13: Cost per run (USD), API calls, token usage, and failure rates. Costs estimated from OpenRouter pricing.

Table 14: GPT-OSS $\mathcal{A}_{\mathrm{H}}$ per-run extremes: best run (fewest failures) vs. worst run (most failures).

	Best Run	Worst Run
Failed scenarios	20	26
Kendall's τ	0.906	0.886
Overall MAE	0.039	0.054
Top-1 Accuracy (%)	93.3	90.8

4.8. Run-to-Run Variance

The results above are five-run averages. To assess trial-to-trial stability, we computed every metric independently for each of the five runs per architecture per model. Figures 9–H.12 show the distributions.

RMSE, which penalizes large errors more heavily than MAE, shows the same architecture separation. $\mathcal{A}_{\mathrm{H}}$ RMSE is < 0.11 for every model and every run, while $\mathcal{A}_{\mathrm{D}}$ never drops below 0.29. The Qwen $\mathcal{A}_{\mathrm{H}}$ box shows a wider interquartile range (RMSE $\in [0.157, 0.166]$), reflecting the GPM estimation imprecision that also limits its τ. The architecture ordering $\mathcal{A}_{\mathrm{H}} > \mathcal{A}_{\mathrm{E}} > \mathcal{A}_{\mathrm{D}}$ holds in every single run across all four models with no overlap: no run of any architecture overtakes another's median metric. This perfect separation confirms that the ranking advantage is not driven by lucky seeds or favourable random initialisations but is a stable structural property of the architecture design. Full per-metric distributions (MAE, RMSE, Top-1 accuracy) appear in Appendix H.

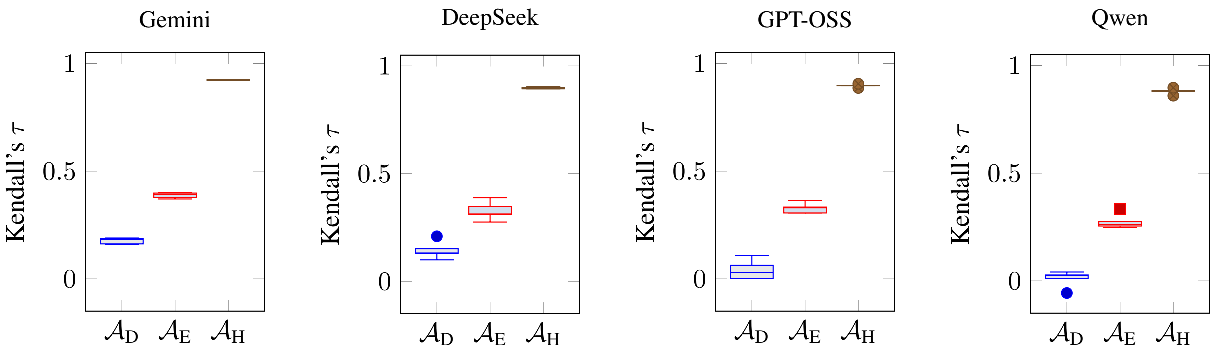

Figure 9: Distribution of Kendall's τ across five independent runs per model. Box plots show median, quartiles, and whiskers.

5. Discussion

5.1. Interpretation of Results

The architecture ordering established in Section 4 is interpreted below.

The model-separation finding deserves emphasis. Across four models spanning two orders of magnitude in parameter count and API cost, $\mathcal{A}_{\mathrm{H}}$ τ varies only between 0.877 and 0.925—a range of 0.048. For $\mathcal{A}_{\mathrm{D}}$, the same spread across models is 0.171; for $\mathcal{A}_{\mathrm{E}}$, 0.096. The $\mathcal{A}_{\mathrm{H}}$ architecture compresses the capability gap: cheaper models become viable in production with minimal accuracy loss. A household-energy deployment using Qwen 3.5 9B through $\mathcal{A}_{\mathrm{H}}$ would achieve 90.3% Top-1 accuracy and $\tau = 0.877$, compared to 93.3% and $\tau = 0.925$ for Gemini 3.5 Flash, at a fraction of the per-token cost.

GPT-OSS-20B had a 12.8% extraction failure rate on $\mathcal{A}_{\mathrm{H}}$ (all recovered by cross-run averaging), yet its $\mathcal{A}_{\mathrm{H}}$ τ of 0.897 was closer to Gemini's 0.925 than to GPT-OSS's own $\mathcal{A}_{\mathrm{D}}$ score of 0.134. Model generation and training quality, not parameter count, explain the 0.048 τ spread across the 9B-to-flagship range.

The ranking advantage of $\mathcal{A}_{\mathrm{H}}$ is partly structural (see Section 3.7). The MAE results confirm that extraction is genuinely accurate: $\mathcal{A}_{\mathrm{H}}$ achieves overall MAE of 0.043–0.068 across models (see Section 4.3), while RMSE remains below 0.11 for all models.

The RAG ablation study (Table 12) confirms that exemplar scores provide the anchoring signal: removing scores drops τ from 0.316 to 0.109, while removing hidden engineering parameters has negligible effect ($\tau = 0.300$ versus 0.316). The near-equivalence of the top cluster (control, retrieval variants, exemplar-content, and alternate embedding; pairwise $p_{\mathrm{Holm}} > 0.97$) and the poor performance of random exemplars ($\tau = 0.062$) together demonstrate that the RAG contribution comes from the semantic selection of similar exemplars whose scores serve as calibration anchors, not from the retrieval architecture or the engineering parameters within the exemplars.

The Top-2 accuracy figures add practical context. $\mathcal{A}_{\mathrm{E}}$ achieves Top-2 of roughly 80% across models, meaning the ground-truth best alternative appears in the architecture's top two about four times out of five. For many household decisions, this is sufficient: if the cost and comfort difference between the top two alternatives is small, the user can make an informed choice from either. The practical sufficiency of Top-2 depends on the decision domain—when the cost gap between first and second place is large, Top-1 accuracy matters; when the alternatives are closely matched, Top-2 provides actionable guidance. $\mathcal{A}_{\mathrm{H}}$ at 97.8% Top-2 effectively eliminates the problem: the best alternative is almost always in the recommended pair.

Three objective weight-validation methods confirm the a priori weights. Entropy weights assign 0.371 to the Environmental criterion (a priori: 0.350), 0.163 to Comfort (a priori: 0.200), and 0.198 to Practicality (a priori: 0.150). Both differences fall within the bounds tested by the sensitivity analysis (Table 11). MEREC weights assign 0.404 to Comfort, the criterion where alternatives diverge most within each scenario (zero-variance occurred in 0% of Comfort scenarios versus 13–18% for other criteria). Implied weights collapse to corner solutions: all weight on Comfort or Practicality, because comfort variance alone explains most pairwise preferences when other criteria show zero variance in 13–45% of scenarios. These data-driven methods cannot capture the normative priorities embedded in the design weights (environmental impact, consumer cost responsiveness), which encode values beyond what the physics alone discriminates.

The $\mathcal{A}_{\mathrm{H}}$ design is not energy-specific. It applies to any domain with a verifiable ground-truth function: insurance triage, vendor scoring, portfolio risk, agricultural cost-benefit calculations. The LLM estimates input parameters and the calculator produces scores, sidestepping uncalibrated LLM scoring entirely.

Households contain multiple occupants with conflicting preferences: different thermostat setpoints, shower durations, appliance schedules. The current framework assumes a single decision-maker with fixed weights. $\mathcal{A}_{\mathrm{H}}$'s cheap single-call extraction makes it practical to run the same scenario under multiple weight vectors and surface disagreement, tying back to the weight-sensitivity analysis in Section 4.5.

5.2. Comparison with Prior Work

These results extend several lines of prior research. Svoboda and Lande (2024) used GPT-4 to construct an AHP decision model for cybersecurity and reported a consistency ratio below the conventional 0.10 threshold, demonstrating that LLMs above a certain capability can produce coherent MCDA structures [108]. Our study generalizes this finding across three integration architectures and four models, showing that the architecture choice—not just model capability—determines whether the LLM output converges to a reliable ranking. The $\mathcal{A}_{\mathrm{D}}$ results reinforce the earlier finding that weaker integration strategies produce unreliable rankings even with capable models.

Shekhovtsov et al. (2026) integrated LLMs with the COMET method for automated MCDA, reporting that structured output formats improved ranking consistency [103]. Our $\mathcal{A}_{\mathrm{H}}$ architecture takes this principle further by replacing LLM scoring with a deterministic calculator altogether, reserving the LLM for parameter extraction only. The resulting $\tau = 0.902$ across 195 scenarios represents a meaningful advance over pure LLM MCDA pipelines.

The finding that $\mathcal{A}_{\mathrm{D}}$ scores cluster in a narrow middle band—a central-tendency bias—aligns with the LLM-as-judge literature. Johnson and Zhang (2024) found GPT-4o essay scores clustering below human ratings with reduced variance [28]; Poličar et al. (2025) documented model-specific leniency profiles relative to human graders [29]. The $\mathcal{A}_{\mathrm{D}}$ results extend this observation to MCDA scoring: without external grounding, the LLM avoids extreme scores, missing the sharp nonlinearities in the ground-truth value functions. Li et al. (2026) documented persistent scoring-bias distortions in LLM-as-a-judge pipelines [26]; our results confirm that these distortions propagate into ranking error when the LLM scores directly.

The $\mathcal{A}_{\mathrm{E}}$ architecture follows the Case-Based Reasoning paradigm described by Aamodt and Plaza (1994), where retrieved prior cases shape the solution [22]. Wiratunga et al. (2024) and Minor and Kaucher (2024) demonstrated that retrieved exemplars improve LLM output quality in legal and business-process domains [23, 24]. Our results extend this finding to MCDA: providing pre-scored exemplars improves τ by 0.19–0.28 over $\mathcal{A}_{\mathrm{D}}$, though the improvement remains well below the $\mathcal{A}_{\mathrm{H}}$ threshold.

The study also speaks to the limits of LLM numeracy documented by Cobbe et al. (2021) and Imani et al. (2023) [19, 20]. Rather than improving the LLM's arithmetic capability—a strategy that has yielded incremental gains—$\mathcal{A}_{\mathrm{H}}$ sidesteps numeracy constraints entirely by delegating computation to a deterministic calculator. This design echoes the tool-augmented LLM approach of Schick et al. (2023) and the program-aided reasoning of Gao et al. (2023), who showed that delegating arithmetic to external tools improves accuracy [109, 110].

In the energy decision-support literature, Dietz et al. (2009) demonstrated that household behavioral changes could cut U.S. emissions by 123 MtC annually if adoption barriers were addressed [4]. Attari et al. (2010) found that consumers systematically misestimate energy use and savings [12]. These findings motivate a tool that bridges consumers' information gap without imposing the MCDA workflow tax described by Leclerc et al. (1995) and Ariely and Zakay (2001) [16, 15]. The $\mathcal{A}_{\mathrm{H}}$ architecture, by accepting natural-language descriptions and returning ranked recommendations, addresses both the information gap and the workflow barrier simultaneously.

5.3. Limitations of the Study

Results are specific to Pennsylvania's grid and utility rate structure. The PJM marginal emissions factors (peak: 1.041 lbs CO_2/kWh; off-peak: 0.976 lbs CO_2/kWh) and Pennsylvania time-of-use rate tiers do not generalize to high-renewables grids, where lower and more variable emissions intensities would compress environmental score distributions. The architecture ordering is large enough that it would likely persist, but the absolute MAE values for environmental and energy-cost criteria would shift.

The environmental criterion for shower decisions uses water consumption in gallons rather than CO_2 emissions. HVAC and Appliance environmental scores reflect grid emissions; Shower environmental scores reflect water volume.

Cross-type comparisons of environmental MAE should account for this distinction; within-type ranking validity is unaffected.

5.3.1. Calculator Simplifications

Five engineering imputations in the ground-truth calculators introduce systematic error. Each affects all architectures equally and does not alter the within-type ranking comparison, but each bounds the absolute fidelity of the ground truth.

The shower calculator interpolates mains water temperature from outdoor temperature using a linear model ($T_{\text{inlet}} = 45\,°\text{F}$ when $T_{\text{out}} \leq 32\,°\text{F}$, $T_{\text{inlet}} = 65\,°\text{F}$ when $T_{\text{out}} \geq 75\,°\text{F}$, linear between). At an outdoor temperature of 45 °F in March, the model yields $T_{\text{inlet}} = 50\,°\text{F}$; the NREL hot-water model gives 48 °F. The 2 °F gap shifts the mixing fraction by roughly 0.015, changing shower energy cost by about $0.003 per shower, and compresses the energy-cost distribution in shoulder seasons.

The appliance budget penalty uses fixed monthly cycle rates (18 for dishwashers, 25 for washers, 24 for dryers) [95, 96, 97]. A household running the dishwasher twice daily (60 cycles/month) faces a monthly cost the calculator underestimates by a factor of three. The budget penalty $P(u)$ activates when utilization $u = C_{\text{monthly}}/B$ exceeds 0.80; a household with high actual usage but low assumed rate may receive no penalty when one is warranted.

The shower practicality model assumes 80% of nominal tank size S is available: $H_{\text{avail}} = 0.8\,S$. Actual draw fraction depends on tank recovery rate, inlet temperature, and heater power. A 0.8 factor may trigger the capacity penalty for a four-person household when a 0.9 factor would not, potentially flipping the ranking between a 7-minute and 9-minute shower alternative.

Insulation tiers (Poor/Medium/Good) correspond to R-value ranges (Poor: R-8 to R-12, Medium: R-13 to R-18, Good: R-19 to R-24), with SEER varying independently (8–22). Appliance age bands map to kWh/cycle ranges that also vary within bands. A scenario with R-15 walls retains R-15 in both the ground truth and the $\mathcal{A}_{\text{H}}$ extraction target, while a scenario with R-8 walls (the low end of the Poor tier) retains R-8. The banding compresses continuous parameters into discrete bins, introducing quantization error that propagates through the value functions. Both pipelines see the same quantized input, so the architecture comparison remains valid, but the absolute scores diverge from what a continuous parameterization would produce.

The HVAC calculator maps occupancy context to a run-time fraction (1.0 for fully occupied, 0.75 for overnight, $1 - 0.5(h_{\text{away}}/24)$ for daytime unoccupied). The mapping discards intra-day variation: a household cycling HVAC off for 2 hours at noon and on for 2 hours at midnight receives the same fraction as one holding steady. The resulting energy use differs by 5–12%.

These five imputations introduce systematic error that affects all architectures equally. The absolute MAE and RMSE values reflect the combined error of LLM extraction and calculator imputation. A perfect LLM extractor would still produce non-zero MAE if the ground truth deviates from measured household data.

A related concern is that the same researcher designed both the $\mathcal{A}_{\text{H}}$ parameter-extraction schema and the ground-truth calculators' input schemas, raising a potential circular-validation risk. We address this by noting that the extraction schema—which targets homeowner-accessible quantities such as insulation tier, flow-rate label, and appliance age band—was frozen before calculator development began. The two schemas are structurally distinct: the LLM estimates qualitative, proxy parameters from scenario text, whereas the calculators consume continuous, engineering-grade values (R-value, GPM, kWh/cycle). The LLM never observes the calculator's internal mapping from proxy to engineering parameter, so its extraction is not conditioned on the scoring function. This single-author pipeline is nonetheless a limitation, and independent schema review would strengthen future work.

Our $\mathcal{A}_{\text{D}}$ and $\mathcal{A}_{\text{E}}$ results each reflect one fixed prompt design; prompt sensitivity was never systematically varied, so part of the architecture gap could reflect prompt quality rather than pure architecture.

A more fundamental constraint concerns the implementation cost of the $\mathcal{A}_{\text{H}}$ architecture itself. The three ground-truth calculators developed for this study—covering HVAC load estimation, appliance TOU scheduling, and shower thermodynamics—each required extensive domain research, iterative calibration, and verification against published standards (ASHRAE, ENERGY STAR, EPA eGRID). In practice, this engineering effort constituted the majority of the total project time. This cost does not diminish $\mathcal{A}_{\text{H}}$'s accuracy advantage, but it does bound its practical applicability: $\mathcal{A}_{\text{H}}$ is the appropriate architecture when a vetted, domain-specific calculator already exists or is worth building for the decision at hand. For decision domains where no such calculator exists and the cost of developing one is prohibitive, $\mathcal{A}_{\text{E}}$ or sufficiently capable $\mathcal{A}_{\text{D}}$ may remain the only accessible options, even at a known accuracy cost.

This tradeoff is the primary practical implication of RQ2 and should be weighed carefully before adopting the $\mathcal{A}_{\mathrm{H}}$ approach in new domains.

A further limitation concerns the choice sets supplied to the architectures. In this benchmark, each scenario presents three specific alternatives that a household might compare. In practice, households rarely know three specific alternatives they want to evaluate; the decision space is continuous and often unstructured. However, the three-alternative framing captures the core ranking task: the LLM must identify which option best satisfies the household's criteria. In most cases, three bands around the same times as the specific alternatives programmed in our test scenarios would produce similar overall results, because the value functions are smooth over the relevant ranges. The benchmark tests ranking accuracy, not choice generation; extending the framework to open-ended alternative generation would be a natural next step.

5.4. Future Work

This study evaluates scoring accuracy only; whether households would actually adopt an LLM-MCDA tool in practice—and the behavioral and workflow factors that govern adoption—is outside its scope and left to future work. AI-assisted decision frameworks have been deployed in operations [111], marketing [112], and clinical settings [113], suggesting that LLM integration into structured decision pipelines is feasible at scale, but the household context introduces unique adoption barriers that remain to be studied.

A prompt-variant robustness check—systematically varying system prompts across architectures—would isolate architecture effects from prompt-quality effects.

Replacing the water-consumption proxy in the Shower environmental criterion with a lifecycle CO_2 factor would make the environmental criterion semantically consistent across all three decision types and enable valid cross-type environmental MAE comparisons. Validating the calculators against instrumented households would quantify the minimum achievable MAE for each architecture.

Several further research directions emerge from this study. First, the criterion weights were supplied as fixed values; a multi-turn conversational interface that iteratively elicits and refines a household's preferences would reduce the cognitive burden of single-shot weight specification and improve the ecological validity of the tool [36, 49]. Such an elicitation loop could use partial rankings or pairwise trade-off questions, grounding the dialogue in the same MAVT framework that underpins the scoring architecture.

Second, the benchmark is confined to the Pennsylvania/PJM grid with its specific rate structure and marginal emissions factors. Cross-grid generalization—evaluating on utilities with different time-of-use tariffs, seasonal pricing, and regional grid mix—would test whether the architecture ordering is a local artifact or a robust structural finding [40, 10]. Cross-country generalization is equally important: residential energy intensity varies widely across nations (from 7 koe/m^2 in China to 18 koe/m^2 in the EU), and HVAC dominates residential consumption differently depending on climate and building stock [1]. Variants of the $\mathcal{A}_{\mathrm{H}}$ calculators that accept grid-specific parameters as inputs rather than hard-coding them would also reduce the per-domain engineering cost identified as a limitation of $\mathcal{A}_{\mathrm{H}}$.

Third, the $\mathcal{A}_{\mathrm{H}}$ architecture currently routes the LLM's extracted parameters to an embedded Python calculator. An agentic design in which the LLM issues structured function calls to external calculators via tool-use APIs would decouple model updates from calculator maintenance, allow runtime selection among candidate calculators, and enable the LLM to request additional information when extraction confidence is low [109, 110, 107].

Fourth, the current appliance scenarios assume a flat electricity rate. Integrating real-time time-of-use pricing and demand-response signals into the scheduling criterion would align the appliance calculator with the dynamic pricing environments that residential consumers increasingly face [59, 38, 81]. This extension would require coupling the appliance cost model to a live or forecasted TOU schedule, which in turn motivates the agentic tool-use design discussed above.

Fifth, all three architectures accept LLM outputs without constraint checking. A neurosymbolic verification layer that checks extracted or scored outputs against known physical bounds—for example, rejecting a shower duration that implies negative water heating energy—would provide a formal safety net that is independent of model capability [114, 115, 116]. Such a layer would be especially valuable for $\mathcal{A}_{\mathrm{D}}$ and $\mathcal{A}_{\mathrm{E}}$, where the LLM has direct scoring authority and no calculator to constrain its outputs.

Sixth, the $\mathcal{A}_{\mathrm{D}}$ and $\mathcal{A}_{\mathrm{E}}$ architectures exhibit known susceptibility to anchoring and position effects in multi-criteria scoring [26, 25]. MCDA debiasing techniques—including prompt-level interventions (randomized criterion ordering, explicit counter-anchoring instructions) and post-hoc normalization against the value-function scale—could narrow the accuracy gap between these architectures and $\mathcal{A}_{\mathrm{H}}$ without requiring a physics calculator [106, 27].

Seventh, the gap between a correct recommendation and an adopted behavior is well documented in the energy-efficiency literature [117, 4]. Combining the LLM-MCDA output with behavioral nudge design—such as defaulting to the recommended setpoint, framing savings as daily rather than annual amounts, or integrating habit-formation prompts—would close the loop between decision quality and real-world energy reduction [50, 118, 77]. Prior work on workflow costs [17] and judgment under time pressure [119] suggests that even accurate tools face adoption barriers when they impose structured input demands on time-constrained users. This direction would require a user study that measures both recommendation fidelity and post-intervention energy consumption.

Finally, a systematic cost-latency-accuracy Pareto analysis across model sizes, inference strategies (single-shot versus chain-of-thought [105]), and caching schemes would quantify the three-way tradeoff that governs deployment viability. Smaller, faster models are cheaper per query but may sacrifice extraction fidelity; agentic tool-use adds latency but offloads computation to deterministic code. Mapping this design space would provide practitioners with the information needed to select an architecture appropriate to their latency budget and accuracy requirements [18, 112]. Taken together, these directions point toward an LLM-MCDA system that is not only accurate on benchmark scenarios but also generalizable, verifiable, behaviorally informed, and deployable at household scale.

6. Conclusion

This study benchmarked three strategies for integrating an LLM into an MCDA workflow for household energy decisions—direct $\mathcal{A}_{\mathrm{D}}$, $\mathcal{A}_{\mathrm{E}}$ prompting, and $\mathcal{A}_{\mathrm{H}}$ that uses the LLM only to extract withheld engineering parameters and a deterministic physics calculator to score—against a physics-based MAVT ground truth across HVAC, Appliance, and Shower decisions. $\mathcal{A}_{\mathrm{H}}$ achieved Kendall's τ of 0.902 and MAE of 0.051, compared to 0.339 and 0.138 for $\mathcal{A}_{\mathrm{E}}$ and 0.095 and 0.219 for $\mathcal{A}_{\mathrm{D}}$. The architecture ordering held across all four models. In answer to RQ1, $\mathcal{A}_{\mathrm{H}}$ most accurately replicates the physics-based ground truth; in answer to RQ2, this ordering held across all four models tested, though the gap between architectures narrowed for weaker models.

Grounding the LLM's numeric reasoning in a deterministic calculator improves agreement with the physics ground truth, and the advantage holds as model capability decreases. A physics backbone compensates for weaker models. We separate this claim into its ranking and score-level components. The ranking robustness of the $\mathcal{A}_{\mathrm{H}}$ design is partly structural (see Section 3.7), so the score-level error and the appliance-timing result are the load-bearing measures of extraction fidelity. The two principal threats—one researcher designed both the calculator and the $\mathcal{A}_{\mathrm{H}}$ extractor, and the single-grid (Pennsylvania/PJM) specificity of the ground truth—are stated in Section 5.3 and bound the generalization. For households, the practical payoff is a decision tool that works from a natural-language description without requiring MCDA expertise: describe the situation, receive a ranked recommendation backed by transparent physics. For grid planners and demand-response programs, the $\mathcal{A}_{\mathrm{H}}$ architecture demonstrates that LLM-assisted MCDA can scale to millions of household decisions—each scored consistently against the same engineering standards—without the per-scenario expert effort that traditional MCDA demands. What's more, creating a calculator, while not feasible on a short notice or for a small set of decisions (which is where $\mathcal{A}_{\mathrm{D}}$ and $\mathcal{A}_{\mathrm{E}}$ shine), can be done for any decision making field. This can lead to an extremely powerful, accessible tool for optimal decision making.

Declaration of competing interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement. **Ahaan Nigam:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing — Original Draft, Writing — Review & Editing, Visualization. **River Huang:** Validation, Conceptualization, Methodology, Supervision, Writing — Review & Editing.

Data availability. All code, prompts, ground-truth calculators, raw run outputs, and analysis scripts are available at `https://github.com/AN00927/LLM-MCDA-Paper`.

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Symbol	Definition
τ	Kendall's rank correlation coefficient
w_i	weight of criterion i ($i = 1 \ldots 4$), summing to 1
$v_i(x)$	value function score for criterion i at raw value x, range $[0, 10]$
α	shape parameter for logarithmic value functions
s_j	MAVT weighted sum score for alternative j
MAE	Mean Absolute Error on MAVT score (0–1 scale)
RMSE	Root Mean Squared Error on MAVT score (0–1 scale)
n	number of test scenarios evaluated
k	number of retrieved chunks per query in RAG ($k = 3$)

Table A.15: Notation used throughout this paper.

Table B.16: Weight comparison: a priori (subjective) vs. objective derivation methods. Entropy and MEREC weights are computed over the full 285-scenario corpus; implied weights from pairwise regression (855 comparisons). The a priori vector $(0.30, 0.35, 0.20, 0.15)$ is the design choice defended in Section 3.3.

Method	Ene	Env	Com	Pra
A priori (design)	0.300	0.350	0.200	0.150
Entropy (overall)	0.268	0.371	0.163	0.198
Entropy (HVAC)	0.336	0.320	0.269	0.074
Entropy (Appliance)	0.193	0.465	0.124	0.218
Entropy (Shower)	0.334	0.314	0.127	0.226
MEREC (overall)	0.194	0.131	0.404	0.272
MEREC (HVAC)	0.138	0.128	0.663	0.071
MEREC (Appliance)	0.208	0.044	0.292	0.456
MEREC (Shower)	0.248	0.244	0.203	0.305
Implied (overall)	0.000	0.000	1.000	0.000
Implied (HVAC)	0.000	0.000	1.000	0.000
Implied (Appliance)	0.000	0.000	0.000	1.000
Implied (Shower)	0.000	0.000	1.000	0.000

Appendix A. Notation

Appendix B. Weight Validation: Entropy, MEREC, and Implied Weights

Table B.16 compares the a priori subjective weights against three objective weight-derivation methods computed on the full ground-truth corpus (195 test + 90 RAG scenarios, 285 total).

Appendix C. Full Per-Model Results

Tables C.17–C.20 report Kendall's τ, Top-1 accuracy (%), and Top-2 accuracy (%) for each architecture$\times$model$\times$decision-type combination, averaged over five independent runs. Results are imputed: scenarios that failed in all five runs carry the imputed value from cross-run matching.

Appendix D. Full Sensitivity Analysis Results

Tables D.21–D.24 report Kendall's τ for each architecture under the nine weight-perturbation scenarios (Table 11) plus the baseline, evaluated separately for each LLM model.

Table C.17: Per-model results: Overall (195 scenarios). τ = Kendall's τ; T1 = Top-1 accuracy (%); T2 = Top-2 accuracy (%).

Architecture	**Model**	τ	T1	T2
$\mathcal{A}_{\text{D}}$	Gemini	0.169	35.9	69.7
	DeepSeek	0.159	40.0	71.8
	GPT-OSS	0.053	33.9	65.1
	Qwen	−0.002	31.8	64.1
$\mathcal{A}_{\text{E}}$	Gemini	0.378	54.4	82.6
	DeepSeek	0.354	49.7	79.5
	GPT-OSS	0.340	51.3	82.1
	Qwen	0.282	51.3	79.5
$\mathcal{A}_{\text{H}}$	Gemini	0.925	93.3	98.5
	DeepSeek	0.897	91.3	97.4
	GPT-OSS	0.901	91.8	99.0
	Qwen	0.877	90.3	96.4

Table C.18: Per-model results: HVAC (70 scenarios).

Architecture	**Model**	τ	T1	T2
$\mathcal{A}_{\text{D}}$	Gemini	−0.048	24.3	62.9
	DeepSeek	−0.124	17.1	67.1
	GPT-OSS	0.238	48.6	78.6
	Qwen	0.057	40.0	67.1
$\mathcal{A}_{\text{E}}$	Gemini	−0.076	18.6	67.1
	DeepSeek	0.048	27.1	61.4
	GPT-OSS	0.267	44.3	80.0
	Qwen	0.190	44.3	72.9
$\mathcal{A}_{\text{H}}$	Gemini	0.981	97.1	100.0
	DeepSeek	0.933	92.9	98.6
	GPT-OSS	0.933	92.9	100.0
	Qwen	0.876	85.7	97.1

Appendix E. RAG Corpus Description

The retrieval-augmented generation (RAG) corpus consists of 90 unique scenarios (35 HVAC, 35 Appliance, 20 Shower) that are fully disjoint from the 195 test scenarios. Each RAG scenario has three alternatives, yielding 270 total rows across the three source files (HVACRagScenarios.xlsx, ApplianceRagScenarios.xlsx, ShowerRagScenarios.xlsx).

Scenarios were generated from the same parameter distribution as the test set but kept separate to prevent any test scenario from appearing in the retrieval index. A combinatorial audit ensured that no parameter combination was underrepresented: each categorical cross-cell (insulation tier $\times$ housing type, appliance type $\times$ flow-rate band, etc.) contained at least one scenario, and the 90 RAG scenarios were drawn from this audited matrix.

The retrieval index was built using the `sentence-transformers/all-MiniLM-L6-v2` embedding model [120], producing 384-dimensional dense vectors. At query time, the three most similar scenarios ($k = 3$) are retrieved by cosine similarity and rendered as full worked examples in the LLM prompt. The embedding string includes all homeowner-visible parameters (location, square footage, insulation tier, household size, housing type, utility budget, outdoor temperature, house age) plus band labels for age and flow rate, but excludes engineering parameters and ground-truth scores, which appear only in the display metadata of the retrieved exemplars.

The disjointness of test and RAG sets was verified by construction (the rebuild script exports them from separate master-sheet partitions) and is enforced at query time by never querying the index with test-scenario embeddings.

Table C.19: Per-model results: Appliance (65 scenarios).

Architecture	Model	τ	T1	T2
$\mathcal{A}_{\mathrm{D}}$	Gemini	-0.026	23.1	58.5
	DeepSeek	-0.056	32.3	55.4
	GPT-OSS	-0.190	26.2	49.2
	Qwen	-0.128	21.5	58.5
$\mathcal{A}_{\mathrm{E}}$	Gemini	0.456	69.2	84.6
	DeepSeek	0.364	56.9	83.1
	GPT-OSS	0.067	38.5	72.3
	Qwen	0.241	50.8	83.1
$\mathcal{A}_{\mathrm{H}}$	Gemini	0.959	96.9	100.0
	DeepSeek	0.959	98.5	100.0
	GPT-OSS	0.959	98.5	100.0
	Qwen	0.897	93.9	96.9

Table C.20: Per-model results: Shower (60 scenarios).

Architecture	Model	τ	T1	T2
$\mathcal{A}_{\mathrm{D}}$	Gemini	0.633	63.3	90.0
	DeepSeek	0.722	75.0	95.0
	GPT-OSS	0.100	25.0	66.7
	Qwen	0.067	33.3	66.7
$\mathcal{A}_{\mathrm{E}}$	Gemini	0.822	80.0	98.3
	DeepSeek	0.700	68.3	96.7
	GPT-OSS	0.722	73.3	95.0
	Qwen	0.433	60.0	83.3
$\mathcal{A}_{\mathrm{H}}$	Gemini	0.822	85.0	95.0
	DeepSeek	0.789	81.7	93.3
	GPT-OSS	0.800	83.3	96.7
	Qwen	0.856	91.7	95.0

Appendix F. Pennsylvania Time-of-Use Rate Structure

Table F.25 lists the six Pennsylvania utilities used by the appliance ground-truth calculator, with their TOU peak/off-peak windows and energy rates ($/kWh). All scenarios use weekday summer rates year-round; seasonal peak-window shifts (e.g., PPL winter 4–8pm vs. summer 2–6pm) and weekend off-peak pricing are not modeled (see Section 3.6 for discussion).

PECO (Rate R-TOU) and PPL (Rate RTS) use a 4-hour afternoon peak window (2–6pm weekdays). The three FirstEnergy utilities (West Penn, Penelec, MetEd) use a 7-hour peak window (2–9pm weekdays), with rates derived from the Jun-2025 Price-to-Compare (PTC) multiplied by approved multipliers [121, 122]. Duquesne uses a flat default rate (Rate RS); an optional TOU schedule (peak 3–9pm) exists but is not applied in the benchmark because the majority of Duquesne customers remain on the flat rate.

Appendix G. GPT-OSS Extraction Failure Example

Appendix G.1. Worked Example: $\mathcal{A}_H$ HVAC Scenario

This subsection traces one HVAC scenario end-to-end through the $\mathcal{A}_{\mathrm{H}}$ pipeline, showing every intermediate value so a reader can verify the computation by hand.

Table D.21: Sensitivity analysis: Kendall's τ under weight perturbations (Gemini 3.5 Flash).

Perturbation	$\mathcal{A}_{\mathrm{D}}$	$\mathcal{A}_{\mathrm{E}}$	$\mathcal{A}_{\mathrm{H}}$
Baseline	0.169	0.378	0.925
Ene $+0.05$	0.200	0.374	0.904
Ene -0.05	0.210	0.443	0.945
Env $+0.05$	0.190	0.385	0.921
Env -0.05	0.217	0.415	0.939
Com $+0.05$	0.210	0.422	0.932
Com -0.05	0.210	0.392	0.901
Pra $+0.05$	0.207	0.415	0.939
Pra -0.05	0.180	0.402	0.915
Equal	0.323	0.463	0.925

Table D.22: Sensitivity analysis: Kendall's τ under weight perturbations (DeepSeek V4 Flash).

Perturbation	$\mathcal{A}_{\mathrm{D}}$	$\mathcal{A}_{\mathrm{E}}$	$\mathcal{A}_{\mathrm{H}}$
Baseline	0.159	0.354	0.897
Ene $+0.05$	0.203	0.357	0.880
Ene -0.05	0.139	0.381	0.939
Env $+0.05$	0.176	0.350	0.891
Env -0.05	0.139	0.364	0.939
Com $+0.05$	0.142	0.388	0.928
Com -0.05	0.217	0.327	0.860
Pra $+0.05$	0.121	0.344	0.918
Pra -0.05	0.197	0.364	0.894
Equal	0.121	0.395	0.921

Step 1: Raw scenario text.. The LLM receives a homeowner-facing question along with known parameters:

- Location: Philadelphia, PA
- Outdoor temperature: 95 °F (cooling season)
- Home size: 2,200 sq ft
- Insulation: Medium
- Household size: 4
- Housing type: Single-family
- House age: 15 years
- Utility budget: $200/month
- Alternatives: thermostat setpoints of 72 °F, 76 °F, and 80 °F

Step 2: $\mathcal{A}_H$ extraction.. The LLM is asked to estimate only the engineering parameters absent from the homeowner description. A typical successful extraction returns:


```
{"decision_type": "HVAC",
 "parameters": {"r_value": 15, "seer": 13,
                "hvac_age": 10,
                "occupancy_context": "occupied_all_day"}}
```

Table D.23: Sensitivity analysis: Kendall's τ under weight perturbations (GPT-OSS 20B).

Perturbation	$\mathcal{A}_{\mathrm{D}}$	$\mathcal{A}_{\mathrm{E}}$	$\mathcal{A}_{\mathrm{H}}$
Baseline	0.053	0.340	0.907
Ene $+0.05$	0.080	0.357	0.904
Ene -0.05	0.074	0.354	0.924
Env $+0.05$	0.050	0.340	0.914
Env -0.05	0.087	0.333	0.931
Com $+0.05$	0.077	0.344	0.918
Com -0.05	0.097	0.306	0.876
Pra $+0.05$	0.077	0.327	0.924
Pra -0.05	0.056	0.371	0.931
Equal	0.142	0.361	0.904

Table D.24: Sensitivity analysis: Kendall's τ under weight perturbations (Qwen 3.5 9B).

Perturbation	$\mathcal{A}_{\mathrm{D}}$	$\mathcal{A}_{\mathrm{E}}$	$\mathcal{A}_{\mathrm{H}}$
Baseline	-0.002	0.282	0.877
Ene $+0.05$	0.039	0.275	0.877
Ene -0.05	-0.019	0.268	0.904
Env $+0.05$	0.009	0.289	0.891
Env -0.05	-0.009	0.251	0.904
Com $+0.05$	-0.029	0.272	0.894
Com -0.05	0.036	0.255	0.833
Pra $+0.05$	-0.002	0.248	0.887
Pra -0.05	0.019	0.306	0.887
Equal	-0.029	0.251	0.901

The validation layer confirms each value is within its physically admissible range ($R \in [11, 19]$ for Medium insulation; SEER $\in [8, 25]$; age ≥ 0; valid occupancy token), so the scenario proceeds to the calculator.

Step 3: Calculator inputs.. The extracted JSON is merged with the known homeowner-facing fields to produce the full scenario dict passed to `HVACGroundTruthCalculator`:

Parameter	**Value**
`outdoor_temp`	95 °F
`square_footage`	2,200
`r_value`	15 (extracted)
`household_size`	4
`housing_type`	Single-family
`seer`	13 (extracted)
`hvac_age`	10 (extracted)
`occupancy_context`	occupied_all_day (extracted)
`utility_budget`	$200
`electricity_rate`	$0.19/kWh
`alternatives`	72, 76, 80

Table F.25: Pennsylvania utility TOU rate structure. Peak window is 24-hour clock (start, end). All rates in \$/kWh. The residential electricity rate of \$0.19/kWh used for HVAC cost normalization is the statewide weighted average [64].

Utility	**Peak Window**	**Peak Rate**	**Off-Peak Rate**
PECO	2–6pm	0.3200	0.0760
PPL	2–6pm	0.1600	0.0700
West Penn	2–9pm	0.1720	0.0880
Penelec	2–9pm	0.1850	0.0930
MetEd	2–9pm	0.2030	0.1000
Duquesne	2–9pm	0.1375	0.1375

Step 4: Thermal load (Eq. 6).. For the 72 °F setpoint ($\Delta T = 95 - 72 = 23\,^{\circ}$F):

$$Q_{\text{cool}} = \underbrace{\tfrac{1.7\times2200}{15} \times 23}_{\text{conductive}} + \underbrace{400(4) + 2200 + 800}_{\text{internal}} + \underbrace{0.15 \times 2200 \times 20}_{\text{solar}} + \underbrace{1.08 \times \tfrac{2200\times8\times0.35}{60} \times 23}_{\text{infiltration}}$$
$$= 5{,}735 + 4{,}600 + 6{,}600 + 2{,}550 = 19{,}485 \text{ BTU/hr}$$

where $m = 1.7$ (single-family multiplier), $R = 15$, $n = 4$, $h_c = 8\,\text{ft}$, $\alpha = 0.35\,\text{ACH}$.

Step 5: EER (Eq. 7)..

$$\text{EER} = -0.02 \times 13^2 + 1.12 \times 13 = -3.38 + 14.56 = 11.18$$

Step 6: Energy, cost, and emissions (Eqs. 8, 4).. With SEER $= 13$, EER $= 11.18$, and $m_{\text{occ}} = 1.0$ (occupied all day):

$$E_{\text{kWh}} = \frac{19{,}485}{11.18 \times 1000} \times 8 \times 1.0 = 13.94 \text{ kWh}$$

$$C = 13.94 \times 0.19 = \$2.65 \qquad \text{CO}_2 = 13.94 \times 1.019 = 14.21 \text{ lbs}$$

where $1.019 = (16 \times 1.041 + 8 \times 0.976)/24$ is the 24-hour peak/off-peak weighted PJM marginal emission factor for occupied-all-day (Eq. 9).

Step 7: Monthly cost and budget penalty (Eq. 5).. The monthly cost proxy is $C_{\text{monthly}} = 2.65 \times 90 = \$238.42$ (three 8-hour periods per day across 30 days). Budget utilization is $u = 238.42/200 = 1.192$, which falls in the $1.0 \le u < 1.5$ exponential-loss-aversion tier:

$$P(u) = 0.5\, e^{-3(1.192-1.0)} = 0.5\, e^{-0.576} = 0.281$$

Step 8: Comfort (Eq. 10).. Cooling season, $T^* = 76\,^{\circ}$F:

$$C_{\text{comfort}} = \tfrac{1}{10}\big(10 - |72 - 76|\big) = \tfrac{1}{10}(6 - 0.4) = 0.56$$

The -0.4 term is the household-size penalty $(4 - 3) \times 0.3 \times 4/3$.

Step 9: Practicality (Eqs. 11–12).. No extremity penalty ($71 < 72 < 82$), so the base score is 10. $\Delta T = 23\,^{\circ}$F gives a multiplier of 0.85. The age/maintenance degradation is $D = \min(0.30,\ 0.010 \times 1.5 \times 10) = 0.150$:

$$P_{\text{prac}} = \frac{10 \times 0.85 \times (1 - 0.150)}{10} = 0.72$$

Step 10: Value-function transformation.. Lower raw cost/emissions are better (decreasing linear VF, Eqs. 2–3); higher comfort/practicality are better (logarithmic VF with $\alpha = 1.5$ and 1.2 respectively). The reference ranges are from Table 5:

Criterion	**Raw**	$v_i(x)$	**Direction**
Energy cost	\$2.65	$(3.29 - 2.65)/(3.29 - 0.38) = 0.22$	lower = better
Environmental	14.21 lbs	$(18.04 - 14.21)/(18.04 - 1.96) = 0.24$	lower = better
Comfort	0.56	$\ln(1 + 1.5 \times 0.56)/\ln(2.5) = 0.67$	higher = better
Practicality	0.72	$\ln(1 + 1.2 \times 0.71)/\ln(2.2) = 0.78$	higher = better

After applying the budget penalty multiplicatively to the energy-cost VF: $v^*_{\text{cost}} = 0.22 \times 0.281 = 0.06$.

Step 11: MAVT weighted sum (Eq. 1)..

$$s_j = 0.30 \times v_{\text{cost}} + 0.35 \times v_{\text{env}} + 0.20 \times v_{\text{comfort}} + 0.15 \times v_{\text{prac}}$$

Step 12: Final ranking.. Repeating Steps 4–11 for all three alternatives yields:

Setpoint	$v^*_{\textbf{cost}}$	$v_{\textbf{env}}$	$v_{\textbf{comfort}}$	$v_{\textbf{prac}}$	**MAVT** s_j
76 °F	0.11	0.30	1.00	0.85	**0.466**
80 °F	0.17	0.37	0.67	0.85	0.441
72 °F	0.06	0.24	0.67	0.78	0.352

The recommended setpoint is 76 °F, which coincides with the ASHRAE 55 comfort optimum—the scenario was designed so the ground-truth calculator's tent-function peak and the budget-constrained energy–comfort tradeoff agree. The 80 °F alternative ranks second despite lower energy cost, because the comfort penalty at four degrees above the optimum outweighs its energy savings under the given weight vector. The 72 °F alternative ranks last: it imposes the highest cost, exceeds the \$200 monthly budget (triggering the exponential tier of $P(u)$), and offers no comfort advantage over 76 °F.

Appendix G.2. GPT-OSS Extraction Failure Analysis

The $\mathcal{A}_{\text{H}}$ architecture validates each extracted JSON before passing it to the ground-truth calculator. A scenario is recorded as an extraction failure when: the LLM response is not valid JSON; a required parameter is absent; a numeric parameter parses as non-finite or falls outside its physically admissible range; or the extracted decision type does not match the scenario's known type.

GPT-OSS 20B exhibited a 12.8% extraction failure rate on $\mathcal{A}_{\text{H}}$ HVAC scenarios, triggered entirely by invalid parameters. In a typical failure the model returns well-formed JSON but estimates parameters outside the valid physical range—for example, returning an R-value of 30 (valid range: 11–19) or a SEER rating of 5 (valid range: 8–25). The validation layer detects the out-of-range value, flags the scenario as an extraction failure, and assigns the sentinel value (1928) to all four criterion scores for that scenario.

Appendix H. Run-to-Run Variance: Full Distributions

Table H.26: Per-model Overall MAE (5-run mean) for each architecture.

Model	$\mathcal{A}_{\text{D}}$	$\mathcal{A}_{\text{E}}$	$\mathcal{A}_{\text{H}}$
Gemini	0.220	0.129	0.047
DeepSeek	0.231	0.141	0.045
GPT-OSS	0.241	0.145	0.052
Qwen	0.235	0.170	0.071

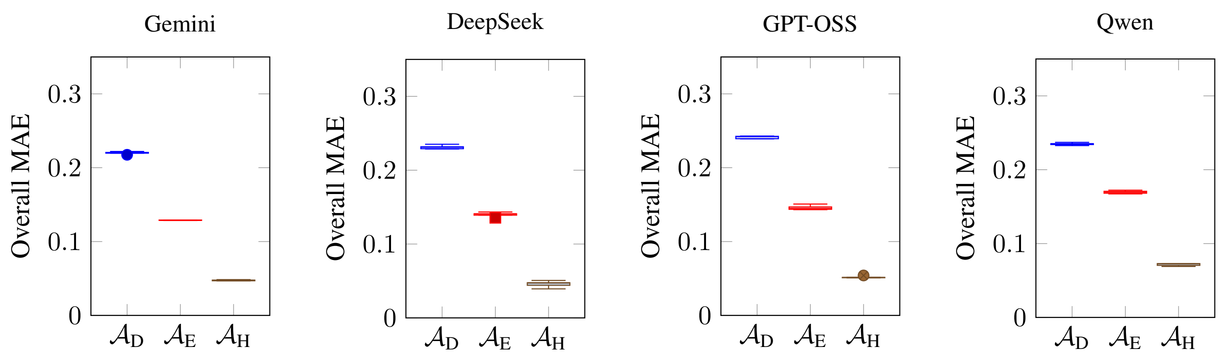

Figure H.10: Distribution of Overall MAE across five independent runs per model.

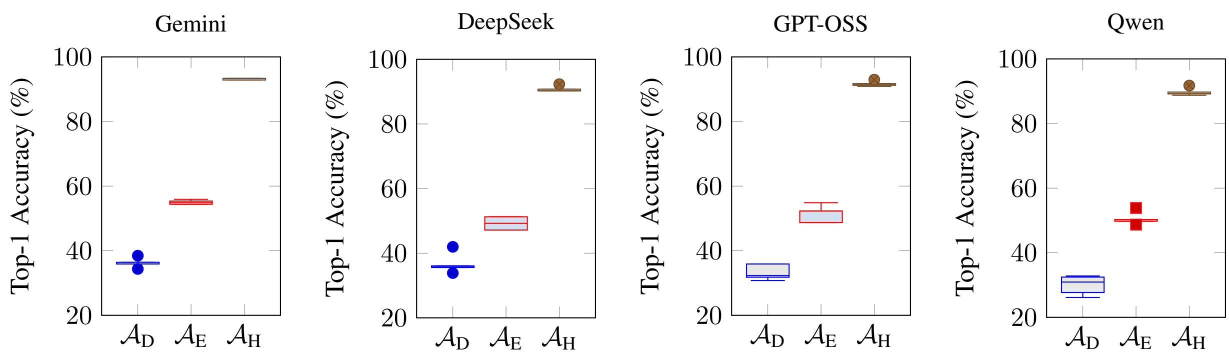

Figure H.11: Distribution of Top-1 Accuracy across five independent runs per model.

Appendix I. Statistical Significance Tests

Three complementary statistical approaches evaluate whether the architecture ordering ($\mathcal{A}_H > \mathcal{A}_E > \mathcal{A}_D$) is statistically robust. All tests use per-scenario metrics computed from the 195 test scenarios, with five independent runs per architecture–model combination.

Appendix I.1. Intraclass Correlation Coefficient

Table I.27 reports ICC(2,1) values decomposing total metric variance into between-model and residual components.

Appendix I.2. Pairwise Wilcoxon Tests with Stouffer Combination

Table I.28 reports Wilcoxon signed-rank statistics for pairwise architecture comparisons on each primary metric, applied independently within each model and combined across models using Stouffer's method. All pairwise comparisons are significant ($p < 0.001$) for every metric.

Appendix I.3. Effect Sizes

Table I.29 reports Cliff's δ for each pairwise comparison.

Appendix I.4. Bootstrap Confidence Intervals

Table I.30 reports 95% bias-corrected and accelerated (BCa) bootstrap confidence intervals for primary metrics, computed over 10,000 resamples of scenarios (with replacement).

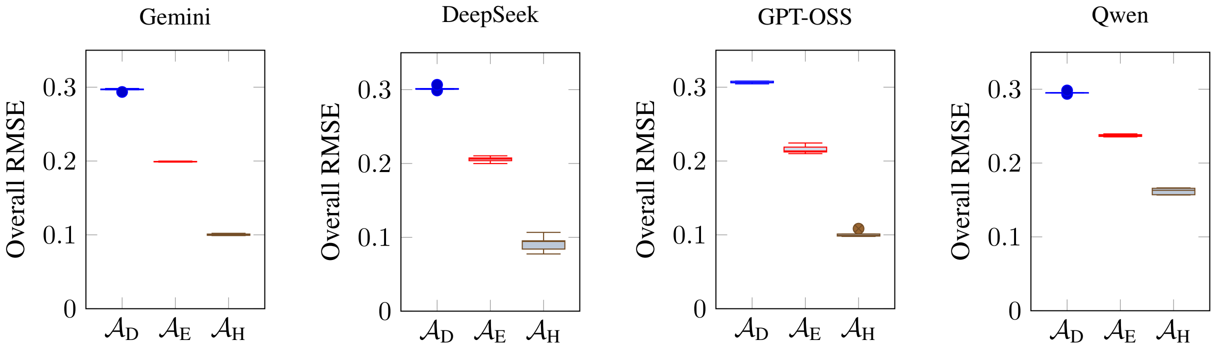

Figure H.12: Distribution of Overall RMSE across five independent runs per model.

Table I.27: ICC(2,1) variance decomposition: fraction of total variance attributable to between-model differences (σ^2_{model}) and residual ($\sigma^2_{\text{residual}}$).

Architecture	Metric	σ^2_{model}	$\sigma^2_{\text{residual}}$
$\mathcal{A}_D$	Kendall's τ	0.427	0.573
	MAE	0.798	0.202
	Top-1	0.081	0.919
$\mathcal{A}_E$	Kendall's τ	0.205	0.795
	MAE	0.914	0.086
	Top-1	0.101	0.899
$\mathcal{A}_H$	Kendall's τ	0.448	0.552
	MAE	0.818	0.182
	Top-1	0.267	0.733

Table I.28: Wilcoxon signed-rank tests: per-model and Stouffer-combined p-values for pairwise architecture comparisons.

		Per-model p-value				
Comparison	Metric	Gemini	DeepSeek	GPT-OSS	Qwen	Stouffer p
$\mathcal{A}_D$ vs. $\mathcal{A}_E$	τ	$<.001$	$<.001$	$<.001$	$<.001$	$<.001$
	MAE	$<.001$	$<.001$	$<.001$	$<.001$	$<.001$
	Top-1	$<.001$	$<.001$	$<.001$	$<.001$	$<.001$
$\mathcal{A}_E$ vs. $\mathcal{A}_H$	τ	$<.001$	$<.001$	$<.001$	$<.001$	$<.001$
	MAE	$<.001$	$<.001$	$<.001$	$<.001$	$<.001$
	Top-1	$<.001$	$<.001$	$<.001$	$<.001$	$<.001$

Table I.29: Cliff's δ effect sizes for pairwise architecture comparisons (mean across four models).

Comparison	τ	MAE	Top-1
$\mathcal{A}_D$ vs. $\mathcal{A}_E$	-0.26 (small)	0.58 (large)	-0.25 (small)
$\mathcal{A}_E$ vs. $\mathcal{A}_H$	-0.69 (large)	0.73 (large)	-0.58 (large)

Table I.30: Bootstrap 95% BCa confidence intervals for primary metrics (all models pooled, 10,000 resamples).

Metric	Architecture	Point Est.	95% CI
Kendall's τ	$\mathcal{A}_{\mathrm{D}}$	0.093	[0.058, 0.125]
	$\mathcal{A}_{\mathrm{E}}$	0.329	[0.307, 0.350]
	$\mathcal{A}_{\mathrm{H}}$	0.899	[0.892, 0.907]
MAE	$\mathcal{A}_{\mathrm{D}}$	0.232	[0.228, 0.235]
	$\mathcal{A}_{\mathrm{E}}$	0.146	[0.140, 0.153]
	$\mathcal{A}_{\mathrm{H}}$	0.054	[0.050, 0.060]
Top-1 (%)	$\mathcal{A}_{\mathrm{D}}$	34.1	[32.5, 35.6]
	$\mathcal{A}_{\mathrm{E}}$	51.5	[50.3, 52.7]
	$\mathcal{A}_{\mathrm{H}}$	91.3	[90.7, 91.9]